

Relational Multiplicity and Fusion-Compatible Internal Gauge Structure in Motion–Staticity Quantum Comparison

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Abstract Previous developments of the motion–staticity (MS) quantum-comparison program identify two structures that are relevant for matter-like sectors but do not yet constitute an internal gauge theory. First, finite rotational comparison may admit a nontrivial projective class characterized by the spin-cover invariant

$$\Xi = -I,$$

which separates spinorial comparison sectors from the tensorial geometric observable algebra. Second, relative-comparison dynamics can, under additional dynamical assumptions, support projective boundary excitations without introducing microscopic spinor variables. Neither result explains how geometry-invisible internal quantum degrees of freedom, a non-Abelian comparison connection, or a common gauge structure acting on several matter sectors should arise.

In this paper we study composites of localized minimal projective comparison carriers and show that their representation multiplicities provide a natural candidate for internal relational degrees of freedom. For

$$\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N},$$

collective rotational comparison gives the decomposition

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j,$$

where V_j carries the geometry-visible spin representation while M_j is invisible to the collective geometric action. Already for three projective carriers one obtains a two-dimensional multiplicity space in the spin- $\frac{1}{2}$ sector, while four carriers contain a two-dimensional rotationally invariant sector. On this neutral relational qubit, pairwise singlet projectors associated with different recouplings fail to commute and generate the full matrix algebra $M_2(\mathbb{C})$. Thus noncommutative internal quantum observables can arise from incompatible relational decompositions even when the total geometric representation is fixed.

We further show that rotationally neutral many-body sectors possess a strong form of geometric blindness: every one-carrier reduced state is maximally mixed, and the projection of any single-carrier vector operator into the neutral sector vanishes. Nevertheless, the dimension of the neutral multiplicity space grows as a Catalan number,

$$\dim \text{Inv}_{SU(2)} [(\mathbb{C}^2)^{\otimes 2n}] = \frac{1}{n+1} \binom{2n}{n},$$

so geometry-neutral relational information remains asymptotically extensive.

When a protected multiplicity space varies from event to event, it defines a vector bundle over spacetime. If $Z(x)$ is an orthonormal local frame and $P(x) = Z(x)Z(x)^\dagger$ the corresponding projector, the induced comparison connection is

$$A = Z^\dagger DZ.$$

A change of basis inside a fixed subspace produces only a pure-gauge connection. Nonzero curvature therefore requires genuine motion of the physical multiplicity subspace,

$$DP \neq 0.$$

The curvature admits the projector form

$$F_A = P(DP \wedge DP)P$$

in a flat ambient comparison bundle, and more generally a Gauss-type decomposition separates ambient curvature from the extrinsic twisting of the protected relational subspace. The maximal frame group $U(m)$ of an m -dimensional multiplicity space is not identified with the physical gauge group; the dynamically realized internal group is instead determined by the holonomy of the relational connection.

Finally, we impose compatibility between inter-event comparison and fusion of localized relational sectors. For an intertwiner

$$C_{rs}^t : M_r \otimes M_s \longrightarrow M_t,$$

transport consistency requires

$$U_\gamma^{(t)} C_{rs}^t = C_{rs}^t (U_\gamma^{(r)} \otimes U_\gamma^{(s)}).$$

This condition leads naturally to a tensor-compatible reconstruction in which the common internal comparison group is associated with tensor automorphisms of the relational fiber functor. In an Abelian sector it immediately yields additive charge selection rules; in non-Abelian sectors it organizes the usual representation-theoretic fusion constraints.

The construction is deliberately kinematical and structural. It does not derive the Standard Model gauge group, chirality, anomaly cancellation, exchange statistics, or a propagating Yang–Mills field. In particular, a Berry or Wilczek–Zee connection on a multiplicity bundle is not by itself a dynamical gauge boson. A genuine gauge-field limit additionally requires an infrared sector with vanishing vector mass and a finite positive transverse kinetic coefficient. The main result is instead a controlled route from projective comparison sectors to geometry-invisible relational multiplicities, from multiplicities to noncommutative internal observables, and from fusion-compatible inter-event transport to candidate internal gauge structure.

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1 Introduction

The quantum extension of the motion–staticity (MS) reconstruction program raises two logically distinct questions. The first is how a classical operational response can become part of a noncommutative quantum theory. The second is how matter-like internal degrees of freedom can arise without being inserted as independent microscopic labels. The former question is addressed by the quantum soldering construction, in which the operational generator–displacement response

$$\Theta_x(P, \xi) = P(e_x \xi)$$

is retained as a commuting configuration observable while coherent deformations of the solder map provide canonically conjugate quantum operators. The latter question remains substantially more open.

The distinction is important. A noncommutative quantum geometry does not by itself produce spinorial matter, and the existence of a spinorial projective sector does not by itself produce an internal gauge symmetry. In the quantum-soldering analysis, finite rotational comparison relative to an appropriately anchored observer frame may carry a nontrivial projective realization of $SO(3)$. The covering-kernel character can be detected by the rephasing-invariant finite-comparison quantity

$$\Xi = U_x(\pi)U_y(\pi)U_x(\pi)^{-1}U_y(\pi)^{-1},$$

for two orthogonal classically commuting π -rotations. In an irreducible spin- j sector,

$$\Xi = (-1)^{2j}I.$$

The nontrivial value

$$\Xi = -I$$

therefore identifies the half-integer projective class. Because the nontrivial element of the spin cover projects to the identity on ordinary Lorentz tensors, the corresponding projective parity lies in the commutant of the tensorial geometric observable algebra. This gives a sharp algebraic distinction between tensorial geometric excitations and spinorial comparison sectors [?].

A complementary relative-comparison construction investigates how such a projective sector might arise dynamically from microscopic variables that are themselves tensorial. If an independent comparison connection is allowed in addition to the geometric Lorentz connection, their relative transport contains rotational modes transforming as $j = 1$ under the local compact little group. Under a suitable hierarchy of comparison modes and a suitable nearest-neighbor interaction, an occupied comparison string can enter a Haldane phase whose open endpoints carry the nontrivial projective class

$$\Xi_{\text{edge}} = -I.$$

This provides a conditional mechanism for projective spin fractionalization without a microscopic spinor variable [?]. The construction is intentionally incomplete: the required channel ordering, deconfinement, localization, continuum covariance, and eventual spin–statistics matching remain additional dynamical questions.

The present paper begins after these two steps. We do not attempt to derive the nontrivial projective sector again. Instead, we ask what new structure becomes available once several localized projective comparison carriers can be combined coherently.

The central observation is that a collection of projective carriers contains more quantum information than is visible in its total geometric representation. Let the minimal nontrivial projective rotational carrier be

$$V_{\frac{1}{2}} \simeq \mathbb{C}^2.$$

For N such carriers,

$$\mathcal{H}_N = \left(V_{\frac{1}{2}}\right)^{\otimes N}. \quad (1)$$

Under the collective $SU(2)$ action this Hilbert space decomposes as

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j. \quad (2)$$

The factor V_j carries the total rotational representation. It is visible to the collective geometric comparison action. The multiplicity factor M_j , by contrast, is acted on trivially by that collective action. It records the different ways in which the same total geometric representation can be assembled from the constituent projective sectors.

This familiar representation-theoretic multiplicity acquires a specific interpretation in the MS setting. The tensorial geometric algebra distinguishes the V_j content but cannot resolve states that differ only inside M_j . The latter therefore provide natural candidates for geometry-invisible relational degrees of freedom. They are not additional spin components and should not be identified with Lorentz indices. Their information concerns how the constituent projective sectors are related while the total geometric representation is held fixed.

The first nontrivial example already occurs for three minimal projective carriers:

$$\left(\frac{1}{2}\right)^{\otimes 3} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}. \quad (3)$$

Equivalently,

$$\mathcal{H}_3 \simeq V_{\frac{3}{2}} \oplus \left(V_{\frac{1}{2}} \otimes \mathbb{C}^2\right). \quad (4)$$

The two-dimensional multiplicity space accompanying total spin $\frac{1}{2}$ is a relational qubit that is invisible to collective rotation.

For four carriers,

$$\left(\frac{1}{2}\right)^{\otimes 4} = 2 \oplus 3(1) \oplus 2(0), \quad (5)$$

or, in multiplicity form,

$$\mathcal{H}_4 \simeq V_2 \oplus \left(V_1 \otimes \mathbb{C}^3\right) \oplus \left(V_0 \otimes \mathbb{C}^2\right). \quad (6)$$

The rotationally invariant subspace is therefore

$$\mathcal{I}_4 := \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes 4} \right] \simeq \mathbb{C}^2. \quad (7)$$

This sector carries no total rotational quantum number at all, yet it contains a complete two-dimensional quantum state space.

The existence of a multiplicity space is not by itself enough to establish nonclassical internal structure. A classical hidden label could also occur with multiplicity. The crucial question is whether physically natural relational observables acting inside the multiplicity space commute.

For two projective carriers i and j , let

$$P_{ij}^{(0)} = \frac{1}{4} (I - \sigma_i \cdot \sigma_j) \quad (8)$$

be the projector onto their rotational singlet. This is a collective rotational scalar,

$$[P_{ij}^{(0)}, \mathbf{J}_{\text{tot}}] = 0,$$

so it preserves the geometric representation sector. Nevertheless, singlet projectors belonging to different recouplings need not commute. Restricted to the four-carrier invariant subspace, one may choose a basis in which

$$P_{12}^{(0)} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{23}^{(0)} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (9)$$

Their commutator is nonzero. In fact, the two relational projectors generate the full matrix algebra

$$M_2(\mathbb{C})$$

on $\mathcal{I}_4$.

This is the first central result of the present work. Quantum noncommutativity can reside entirely in the choice of relational decomposition while every operator involved remains a scalar under the total geometric rotation group. The internal qubit is therefore not introduced as an additional microscopic two-state system. It is generated by incompatible questions about how several projective comparison carriers are coupled to the same total geometric sector.

The same neutral sectors also display a useful form of geometric blindness. If

$$|\Psi\rangle \in \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes N} \right],$$

then the reduced state of every individual carrier is

$$\rho_i = \frac{1}{2} I. \quad (10)$$

No single constituent contains a preferred geometric spin direction. Moreover, if P_I denotes the projector onto the total invariant sector, then every one-carrier vector operator satisfies

$$P_I J_i^a P_I = 0. \quad (11)$$

The leading operators that act nontrivially within the relational sector are therefore composite rotational scalars such as

$$\mathbf{J}_i \cdot \mathbf{J}_j.$$

The logical information is encoded in relations rather than in one-body geometric observables.

This protection does not make the neutral sector small. For $N = 2n$, the dimension of the rotationally invariant multiplicity space is the Catalan number

$$\dim \mathcal{I}_{2n} = C_n = \frac{1}{n+1} \binom{2n}{n}. \quad (12)$$

Hence

$$\log_2 \dim \mathcal{I}_{2n} = 2n - \frac{3}{2} \log_2 n + O(1). \quad (13)$$

A large quantum comparison network can therefore be exactly neutral under collective geometric rotation while retaining an asymptotically extensive amount of internal relational quantum information.

The second question addressed in this paper is how such local multiplicity spaces can be compared between different spacetime events. Suppose a protected m -dimensional relational sector varies smoothly with $x \in M$. Relative to a fixed or covariantly transported ambient comparison space, let

$$Z(x) = (|m_1(x)\rangle, \dots, |m_m(x)\rangle), \quad Z^\dagger Z = I_m, \quad (14)$$

be a local orthonormal frame and define

$$P(x) = Z(x)Z(x)^\dagger. \quad (15)$$

The frame is not unique:

$$Z(x) \mapsto Z(x)U(x), \quad U(x) \in U(m). \quad (16)$$

The induced comparison connection is

$$A = Z^\dagger DZ, \quad (17)$$

where D denotes the chosen ambient covariant derivative. Under Eq. (16), A transforms as a $U(m)$ connection.

This construction must be interpreted carefully. An m -dimensional multiplicity space does not imply that the physical internal gauge group is $U(m)$. If the physical subspace itself is fixed,

$$P(x) = P_0,$$

and only its basis varies,

$$Z(x) = Z_0 U(x),$$

then

$$A = U^{-1} dU$$

up to convention and the curvature vanishes. A nontrivial Berry or Wilczek–Zee curvature therefore requires genuine motion of the protected subspace in the ambient comparison space:

$$DP \neq 0. \quad (18)$$

In a flat ambient bundle its curvature can be written directly in terms of the projector,

$$F_A = P(DP \wedge DP)P, \quad (19)$$

again up to the conventional placement of factors of i . More generally, if

$$K = (I - P)DZ$$

denotes the extrinsic derivative of the protected subspace, the induced curvature obeys a Gauss-type relation of the form

$$F_A = Z^\dagger F_{\text{amb}} Z + K^\dagger \wedge K, \quad (20)$$

with the precise sign determined by the connection convention.

This formulation leads naturally to Grassmannian geometry. A protected m -plane inside an N -dimensional ambient comparison space defines a map

$$M \longrightarrow \text{Gr}(m, N). \quad (21)$$

The universal bundle over the Grassmannian then supplies the natural kinematical setting for relational multiplicity bundles. In the stable limit,

$$\text{Gr}(m, \infty) \simeq BU(m),$$

so sufficiently large comparison spaces possess the topological capacity to support general complex rank- m bundles. This is a statement about kinematical and topological capacity, not a derivation of arbitrary physical gauge fields.

The distinction becomes sharper at the level of holonomy. The full group $U(m)$ is the maximal frame freedom of an m -dimensional complex multiplicity space. The group actually generated by comparison transport is instead

$$G_{\text{hol}} = \overline{\text{Hol}(\nabla_{\text{rel}})} \subseteq U(m). \quad (22)$$

Its Lie algebra is generated by parallel transports of the curvature operators. In the usual finite-dimensional smooth setting this is governed by the Ambrose–Singer construction,

$$\mathfrak{g}_{\text{hol}} = \text{Lie} \left\langle U_\gamma^{-1} F_x(X, Y) U_\gamma \right\rangle. \quad (23)$$

For a two-dimensional multiplicity space, a curvature family confined to one commuting generator produces at most an Abelian holonomy, whereas two noncommuting curvature directions can generate $\mathfrak{su}(2)$. Non-Abelian internal comparison is therefore a curvature-closure property, not a consequence of multiplicity dimension alone.

The third problem is universality across different relational sectors. Separate degenerate subspaces can each carry a Berry connection, but this would merely produce a collection of unrelated comparison connections. A physical gauge principle requires a stronger statement: transport must respect the way sectors combine.

Let $r, s, t, \dots$ label localized relational sectors and let

$$C_{rs}^t : M_r \otimes M_s \longrightarrow M_t \quad (24)$$

denote a nonzero fusion intertwiner. If comparison transport along a path $\gamma : x \rightarrow y$ is represented by

$$U_\gamma^{(r)}, \quad U_\gamma^{(s)}, \quad U_\gamma^{(t)},$$

then fusion and transport should define the same physical comparison whether fusion is performed before or after transport. This gives

$$U_\gamma^{(t)} C_{rs}^t(x) = C_{rs}^t(y) \left(U_\gamma^{(r)} \otimes U_\gamma^{(s)} \right). \quad (25)$$

Infinitesimally this is the covariant constancy condition

$$\nabla C_{rs}^t = 0. \quad (26)$$

Equation (25) changes the interpretation of the internal connection. The different matter sectors are no longer assigned independent comparison gauge fields. Instead, their transports must be different representations of a single comparison structure compatible with the full fusion network.

This viewpoint suggests a Tannaka-type reconstruction. Let

$$\mathcal{C}_{\text{rel}}$$

denote a tensor category of localized, transportable relational sectors and let

$$\omega_x : \mathcal{C}_{\text{rel}} \longrightarrow \text{Hilb}_{\text{fd}} \quad (27)$$

assign to each sector its multiplicity fiber at x . Under the usual rigidity and reconstruction hypotheses, the group of tensor-compatible automorphisms

$$G_x = \text{Aut}^{\otimes}(\omega_x) \quad (28)$$

provides the natural common internal symmetry group. If comparison transport is tensor-natural, the connection seen in a sector r takes the form

$$A^{(r)} = \rho_{r*}(A) \quad (29)$$

for a single principal comparison connection A .

The Tannaka–Krein and Doplicher–Roberts reconstruction theorems used to motivate this structure are standard mathematical results. The claim of the present work is not a new abstract reconstruction theorem. The MS-specific question is whether localized projective comparison sectors, their multiplicity spaces, and their physical fusion operations provide the objects, morphisms, and tensor structure required for such a reconstruction.

The fusion condition also gives an immediate interpretation of charge conservation. In an Abelian sector, write

$$\rho_r(e^{i\theta}) = e^{iq_r\theta}.$$

If

$$C_{rs}^t \neq 0,$$

tensor compatibility requires

$$q_t = q_r + q_s. \quad (30)$$

If

$$r \otimes \bar{r} \longrightarrow \mathbf{1},$$

then

$$q_{\bar{r}} = -q_r.$$

In this sense additive charge conservation is the Abelian form of the requirement that internal comparison preserve the relational fusion network.

The preceding discussion concerns gauge structure, not yet gauge dynamics. This distinction is essential. A geometric or Berry connection on a multiplicity bundle may be entirely composite,

$$A_\mu = A_A(\lambda)\partial_\mu\lambda^A,$$

and need not possess independent propagating degrees of freedom. The appearance of nontrivial holonomy therefore does not by itself establish a Yang–Mills field.

A dynamical gauge-field limit requires the relational comparison connection to become an independent infrared variable. At quadratic order one may characterize the transverse sector schematically by

$$\Gamma_{\mu\nu}^{(2)}(k) = (m_A^2 + K_A k^2 + O(k^4)) P_{\mu\nu}^T + \text{longitudinal terms.} \quad (31)$$

A massless propagating internal comparison field requires

$$m_A^2 = 0, \quad 0 < K_A < \infty. \quad (32)$$

Only in such a regime can the coefficient

$$K_A$$

be interpreted as the inverse low-energy gauge stiffness, conventionally written as g^{-2} . Determining whether a microscopic MS comparison dynamics actually satisfies these conditions is beyond the kinematical analysis developed here.

The logical structure of the paper is therefore deliberately layered. We assume the availability of localized minimal projective comparison carriers and study their composites. Representation multiplicity produces geometry-invisible relational state spaces. Relational scalar observables acting on these spaces are generically noncommuting. Protected multiplicity spaces that vary across spacetime form vector bundles with induced comparison connections. Their physical internal group is determined by curvature and holonomy rather than by multiplicity dimension alone. Compatibility of transport with fusion then supplies the structure required for a common internal comparison group across different sectors. Propagating gauge dynamics remains a further infrared problem.

Several limitations should be stated at the outset. The condition

$$\Xi = -I$$

does not by itself select $j = \frac{1}{2}$; the use of the minimal projective carrier is an additional infrared minimality assumption. Projective spin does not by itself imply Fermi exchange statistics. Localization and deconfinement of the constituent comparison sectors are not derived here. The multiplicity construction does not determine the Standard Model gauge group, the number of generations, chirality, hypercharges, anomaly cancellation, or coupling constants. Nor does a non-Abelian Berry curvature by itself establish a propagating Yang–Mills boson. These distinctions are maintained throughout.

The purpose of the present paper is narrower but more structural. It develops a precise route by which internal quantum information can arise from relations among projective comparison sectors while remaining invisible to the collective geometric action. It then identifies the mathematical conditions under which such relational information can be transported nontrivially between events and consistently across fusion channels. The resulting chain is

$$\begin{aligned} \text{projective comparison carriers} &\longrightarrow \text{representation multiplicity,} \\ \text{representation multiplicity} &\longrightarrow \text{noncommutative relational algebra,} \\ \text{event-dependent protected multiplicity} &\longrightarrow \text{relational connection and holonomy,} \\ \text{fusion-compatible transport} &\longrightarrow \text{common internal comparison structure.} \end{aligned} \quad (33)$$

The remainder of the paper is organized as follows. Section 2 develops the multiplicity decomposition of composite projective comparison sectors and identifies the corresponding geometry-invisible relational state spaces. Section 3 studies the algebra of relational observables, including the four-carrier invariant qubit and the noncommutativity of different singlet recouplings, and analyzes the

scaling and geometric blindness of neutral relational sectors. Section 4 promotes protected multiplicity spaces to spacetime bundles and develops the induced Grassmannian connection, curvature, and holonomy structure, including the distinction between basis gauge freedom and genuine subspace curvature. Section 5 imposes compatibility between comparison transport and sector fusion and formulates the corresponding tensor-compatible reconstruction of a common internal group. Section 6 separates internal gauge kinematics from gauge-field dynamics and states the infrared criteria required for connection liberation. The final section summarizes the logical status of the construction and the dynamical problems that remain.

2 Composite Projective Sectors and Relational Multiplicity

The preceding construction isolates projective comparison sectors by the action of the nontrivial element of the spin covering group. In particular, a sector satisfying

$$\Xi = -I$$

belongs to the half-integer projective rotational class. This condition alone does not select a unique spin representation. In the present paper we are interested in the simplest nontrivial composite problem and therefore begin with the minimal projective carrier.

Assumption 2.1 (Minimal localized projective carrier). *There exists a regime in which localized and coherently composable projective comparison excitations are described, under the anchored rotational comparison group, by the minimal nontrivial irreducible carrier*

$$V_{\frac{1}{2}} \simeq \mathbb{C}^2.$$

The corresponding spin-cover center acts as

$$-I$$

on each such carrier.

Assumption 2.1 is not a consequence of $\Xi = -I$ alone. The projective invariant permits all half-integer spins, and a dynamical mechanism is required to select the lowest carrier, localize it, and make several such excitations coherently composable. The relative comparison construction provides one conditional route to projective spin- $\frac{1}{2}$ boundary excitations, but the present analysis does not depend on the details of that mechanism. We take the resulting minimal carriers as input and study the quantum information contained in their composites.

For N projective carriers, define

$$\mathcal{H}_N := (V_{\frac{1}{2}})^{\otimes N} \simeq (\mathbb{C}^2)^{\otimes N}. \quad (34)$$

Let

$$u_{\frac{1}{2}} : SU(2) \longrightarrow U(V_{\frac{1}{2}})$$

denote the fundamental spin representation. The collective rotational comparison action is

$$U_N(g) = u_{\frac{1}{2}}(g)^{\otimes N}, \quad g \in SU(2). \quad (35)$$

Its infinitesimal generators are

$$J_{\text{tot}}^a = \sum_{i=1}^N J_i^a, \quad J_i^a = I^{\otimes(i-1)} \otimes \frac{\sigma^a}{2} \otimes I^{\otimes(N-i)}. \quad (36)$$

We set $\hbar = 1$ in the purely representation-theoretic formulas of this section. Factors of $\hbar$ may be restored by replacing J_i^a with $\hbar\sigma^a/2$.

The standard decomposition of the tensor product into irreducible collective spin representations takes the form

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}, \quad (37)$$

where V_j is the irreducible spin- j representation and $M_j^{(N)}$ is the corresponding multiplicity space. The allowed values of j are

$$j = \frac{N}{2}, \frac{N}{2} - 1, \dots, j_{\min},$$

with

$$j_{\min} = \begin{cases} 0, & N \text{ even,} \\ \frac{1}{2}, & N \text{ odd.} \end{cases}$$

We write

$$m_j^{(N)} := \dim M_j^{(N)}. \quad (38)$$

The multiplicities have a simple closed form.

Proposition 2.2 (Multiplicity formula). *For N spin- $\frac{1}{2}$ projective carriers,*

$$m_j^{(N)} = \binom{N}{\frac{N}{2} - j} - \binom{N}{\frac{N}{2} - j - 1}, \quad (39)$$

where a binomial coefficient is understood to vanish whenever its lower argument lies outside the interval $0, \dots, N$.

Proof. Let

$$n_M$$

denote the dimension of the subspace with fixed eigenvalue M of the total generator J_{tot}^3 . A basis state with k spins down has

$$M = \frac{N}{2} - k,$$

so

$$n_M = \binom{N}{\frac{N}{2} - M}. \quad (40)$$

Every spin- j irreducible representation contributes exactly one vector to each J_{tot}^3 eigenspace with

$$-j \leq M \leq j.$$

Consequently,

$$n_M = \sum_{j \geq |M|} m_j^{(N)}. \quad (41)$$

Taking

$$M = j$$

and subtracting the corresponding relation at $M = j + 1$ gives

$$m_j^{(N)} = n_j - n_{j+1}.$$

Using Eq. (40) yields Eq. (39). □

The decomposition

$$V_j \otimes M_j^{(N)}$$

contains two conceptually different types of information. The first factor specifies how the composite transforms under collective rotational comparison. The second records how many inequivalent coupling patterns produce the same collective representation.

This distinction motivates the following terminology.

Definition 2.3 (Relational multiplicity space). *For a fixed collective geometric spin sector V_j , the factor*

$$M_j^{(N)}$$

in Eq. (37) will be called the relational multiplicity space. A quantum degree of freedom supported entirely in $M_j^{(N)}$ will be called relationally internal relative to the collective rotational comparison action.

The qualifier “relative to the collective rotational comparison action” is essential. At this stage, relationally internal does not mean invariant under every possible geometric observable in a complete gravitational theory. The statement is more precise: once the total $SU(2)$ representation V_j has been fixed, the collective rotational action cannot distinguish states that differ only in their multiplicity coordinate.

Indeed, Eq. (37) may be chosen so that

$$U_N(g) = \bigoplus_j u_j(g) \otimes I_{M_j^{(N)}}. \quad (42)$$

Thus the collective group acts trivially on every multiplicity factor.

Define the finite-dimensional algebra generated by the collective rotational representation,

$$\mathcal{A}_N^{\text{coll}} := \text{Alg} \{U_N(g) ; g \in SU(2)\} \subseteq \text{End}(\mathcal{H}_N), \quad (43)$$

where the unital complex algebra generated by the representation is intended.

Its commutant identifies the complete finite-dimensional algebra invisible to the collective rotational action.

Theorem 2.4 (Collective commutant decomposition). *Under the decomposition*

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)},$$

the collective rotational algebra and its commutant take the form

$$\mathcal{A}_N^{\text{coll}} \simeq \bigoplus_j \text{End}(V_j) \otimes I_{M_j^{(N)}}, \quad (44)$$

$$\left(\mathcal{A}_N^{\text{coll}}\right)' \simeq \bigoplus_j I_{V_j} \otimes \text{End}\left(M_j^{(N)}\right). \quad (45)$$

Proof. The representation of $SU(2)$ is completely reducible. After grouping equivalent irreducible representations, it has the form Eq. (37), with the collective group acting as Eq. (42). By irreducibility of V_j and Schur's lemma, an operator commuting with every $u_j(g)$ must act as the identity on V_j , up to an arbitrary operator on the associated multiplicity space. Operators cannot mix inequivalent spin sectors while commuting with the full group action. Hence Eq. (45) follows.

Conversely, the bicommutant on each finite-dimensional irreducible block generates the full matrix algebra on V_j , giving Eq. (44). $\square$

The theorem gives a precise algebraic meaning to geometry-invisible multiplicity in the present setting. Once the collective rotational content has been fixed, the remaining matrix algebra

$$\text{End}(M_j^{(N)}) \quad (46)$$

commutes with the entire collective rotational representation.

This result should be compared with the more general commutant structure already suggested by quantum soldering,

$$\mathcal{H} \simeq \bigoplus_{\alpha} V_{\alpha}^{\text{geom}} \otimes M_{\alpha}.$$

The present construction supplies an explicit finite composite mechanism by which nontrivial multiplicity spaces can arise: they need not be postulated as independent microscopic internal labels but can result from repeated composition of minimal projective comparison carriers.

The smallest examples are particularly instructive.

Example 2.5 (Three projective carriers). *For $N = 3$,*

$$\left(V_{\frac{1}{2}}\right)^{\otimes 3} \simeq V_{\frac{3}{2}} \oplus \left(V_{\frac{1}{2}} \otimes \mathbb{C}^2\right). \quad (47)$$

Indeed,

$$m_{\frac{3}{2}}^{(3)} = 1, \quad m_{\frac{1}{2}}^{(3)} = 2.$$

Thus the total spin- $\frac{1}{2}$ sector contains a two-dimensional multiplicity space even though each constituent carries only the minimal spinorial projective representation.

A convenient basis of the multiplicity space may be constructed by choosing an intermediate coupling channel. Couple the first two carriers to

$$j_{12} = 0 \quad \text{or} \quad j_{12} = 1,$$

and then couple the result to the third carrier to obtain total

$$j = \frac{1}{2}.$$

The two admissible values of j_{12} label the two multiplicity states:

$$|0_L\rangle \sim \left| \left[(12)_{j_{12}=0}, 3 \right]_{j=\frac{1}{2}} \right\rangle, \quad |1_L\rangle \sim \left| \left[(12)_{j_{12}=1}, 3 \right]_{j=\frac{1}{2}} \right\rangle. \quad (48)$$

The label L emphasizes that this is a multiplicity or logical coordinate, not an additional physical spin component.

Under collective rotation,

$$|\psi\rangle_{V_{1/2}} \otimes (a|0_L\rangle + b|1_L\rangle)$$

transforms only through the first factor:

$$U_3(g) [|\psi\rangle \otimes |\lambda\rangle] = u_{\frac{1}{2}}(g) |\psi\rangle \otimes |\lambda\rangle. \quad (49)$$

Thus a full qubit degree of freedom is invisible to collective rotational comparison while the total geometric spin remains fixed at $\frac{1}{2}$.

The four-carrier system contains an even sharper example because a nontrivial multiplicity survives after all total spin has been removed.

Example 2.6 (Four projective carriers). *For $N = 4$,*

$$\left(V_{\frac{1}{2}}\right)^{\otimes 4} \simeq V_2 \oplus \left(V_1 \otimes \mathbb{C}^3\right) \oplus \left(V_0 \otimes \mathbb{C}^2\right). \quad (50)$$

The multiplicities are

$$m_2^{(4)} = 1, \quad m_1^{(4)} = 3, \quad m_0^{(4)} = 2.$$

In particular,

$$\mathcal{I}_4 := \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes 4} \right] \simeq \mathbb{C}^2. \quad (51)$$

The space $\mathcal{I}_4$ is rotationally neutral:

$$J_{\text{tot}}^a |\Psi\rangle = 0, \quad |\Psi\rangle \in \mathcal{I}_4. \quad (52)$$

Nevertheless,

$$\dim \mathcal{I}_4 = 2.$$

Thus total geometric neutrality does not imply absence of quantum information.

A useful orthonormal basis can again be chosen by coupling the first pair and the second pair. Let

$$|s_{ij}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle)$$

be the two-carrier singlet. Define

$$|0_L\rangle := |s_{12}\rangle \otimes |s_{34}\rangle. \quad (53)$$

A second invariant state is obtained by coupling the two pairwise triplets to total spin zero:

$$|1_L\rangle := \frac{1}{\sqrt{3}} (|t_{12}^+ t_{34}^- \rangle - |t_{12}^0 t_{34}^0 \rangle + |t_{12}^- t_{34}^+ \rangle), \quad (54)$$

where

$$\begin{aligned} |t_{ij}^+\rangle &= |\uparrow_i \uparrow_j\rangle, \\ |t_{ij}^0\rangle &= \frac{1}{\sqrt{2}} (|\uparrow_i \downarrow_j\rangle + |\downarrow_i \uparrow_j\rangle), \\ |t_{ij}^-\rangle &= |\downarrow_i \downarrow_j\rangle. \end{aligned} \quad (55)$$

Then

$$\{|0_L\rangle, |1_L\rangle\}$$

is an orthonormal basis of $\mathcal{I}_4$.

The subscript L again denotes a relational logical basis. Neither $|0_L\rangle$ nor $|1_L\rangle$ carries a collective rotational orientation. Their distinction concerns the internal recoupling pattern of the same four projective carriers.

The four-carrier example already exhibits the distinction between *geometric neutrality* and *relational triviality*. The former means

$$J_{\text{tot}}^a = 0$$

within the sector. The latter would mean that the sector is one-dimensional and therefore carries no nontrivial internal operator algebra. Equation (51) shows that the two notions are inequivalent.

More generally, for even N , define the neutral relational sector

$$\mathcal{I}_N := \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes N} \right]. \quad (56)$$

For $N = 2n$,

$$\mathcal{I}_{2n} \simeq M_0^{(2n)}.$$

Thus the neutral sector is itself a pure multiplicity space. The collective rotational representation is completely trivial on it:

$$U_{2n}(g)|_{\mathcal{I}_{2n}} = I_{\mathcal{I}_{2n}} \quad \text{for all } g \in SU(2). \quad (57)$$

Consequently, every operator on $\mathcal{I}_{2n}$ commutes with collective rotation after restriction to this sector.

This observation leads to a useful structural statement.

Theorem 2.7 (Existence of geometry-neutral relational quantum sectors). *Let $N = 2n \geq 4$ minimal projective comparison carriers transform collectively under $SU(2)$. Then their total Hilbert space contains a rotationally invariant multiplicity sector*

$$\mathcal{I}_{2n} = \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes 2n} \right]$$

of dimension

$$\dim \mathcal{I}_{2n} = m_0^{(2n)} = \binom{2n}{n} - \binom{2n}{n-1}.$$

Equivalently,

$$\dim \mathcal{I}_{2n} = \frac{1}{n+1} \binom{2n}{n}. \quad (58)$$

For $n \geq 2$, this dimension is greater than one. Hence exact collective rotational neutrality is compatible with nontrivial quantum multiplicity.

Proof. The invariant sector is the total spin-zero sector, whose multiplicity is obtained from Proposition 2.2:

$$m_0^{(2n)} = \binom{2n}{n} - \binom{2n}{n-1}.$$

Using

$$\binom{2n}{n-1} = \frac{n}{n+1} \binom{2n}{n}$$

gives Eq. (58). For $n \geq 2$, the resulting Catalan number is at least two. $\square$

The appearance of the Catalan numbers will be used in the next section to study the information capacity of the neutral relational sector. For the moment the important point is qualitative: imposing exact vanishing of the collective spin does not collapse the composite Hilbert space to a unique state.

At this point one might be tempted to identify the unitary group of the multiplicity space,

$$U(m_j^{(N)}),$$

with an internal gauge group. Such an identification would be premature.

The decomposition

$$V_j \otimes M_j^{(N)}$$

implies that an arbitrary unitary acting on the multiplicity factor,

$$I_{V_j} \otimes U, \quad U \in U(M_j^{(N)}), \quad (59)$$

commutes with collective rotation. It therefore defines a maximal *kinematical* unitary freedom invisible to the collective geometric action. Nothing in the decomposition, however, implies that all such unitaries are physically realizable by local comparison processes, that they can be transported consistently between spacetime events, or that they possess an independent gauge-field dynamics.

We therefore distinguish two structures.

Definition 2.8 (Kinematical commutant group). *For a multiplicity sector $M_j^{(N)}$, define*

$$G_{j,N}^{\text{kin}} := U(M_j^{(N)}). \quad (60)$$

This is the maximal unitary frame group acting on the relational multiplicity space while leaving the collective spin representation fixed.

The adjective “kinematical” is essential. The physical internal comparison group, if one emerges, must be selected by additional structure such as localization, admissible relational observables, inter-event transport, curvature, fusion compatibility, and dynamics.

In particular,

$$\dim M_j^{(N)} = m \quad \not\Rightarrow \quad G_{\text{physical}} = U(m). \quad (61)$$

The multiplicity decomposition determines the largest unitary group that could act invisibly on the collective rotational representation. It does not determine which subgroup is realized by comparison holonomy, whether that subgroup is Abelian or non-Abelian, or whether any associated connection is a propagating gauge field.

A second caution concerns the dependence on the number of constituents. For different values of N , the multiplicities

$$m_j^{(N)}$$

change. Hence the groups

$$U(m_j^{(N)})$$

do not in general define one stable internal group across all composite sectors. A gauge symmetry cannot therefore be reconstructed by selecting one finite- N multiplicity and declaring its full unitary group fundamental. A physically meaningful internal group must act compatibly on a family of sectors and preserve the relations by which those sectors are composed.

This issue will become central when fusion is introduced later in the paper. Before reaching that point, however, we first need to establish that the multiplicity degrees of freedom are genuinely quantum rather than merely degenerate classical labels.

The distinction is nontrivial. The existence of a two-dimensional multiplicity space such as

$$\mathcal{I}_4 \simeq \mathbb{C}^2$$

does not by itself prove that natural relational observables generate the full noncommutative qubit algebra. A preferred set of physical relational questions could, in principle, all be simultaneously diagonalizable, leaving the multiplicity label effectively classical.

The next section addresses precisely this issue. Pairwise neutral-comparison projectors associated with different recoupling channels will be shown to commute with the collective geometric action while failing to commute with one another inside the same multiplicity sector. In the four-carrier neutral space they generate the full algebra

$$M_2(\mathbb{C}).$$

Thus the quantum character of relational multiplicity does not arise merely from the dimension of the commutant; it is witnessed directly by the incompatibility of physically natural relational observables.

3 Noncommutative Relational Observables and Neutral Quantum Information

Section 2 showed that composite minimal projective comparison carriers contain multiplicity spaces

$$M_j^{(N)}$$

on which collective rotational comparison acts trivially. The existence of such a multiplicity space, however, does not by itself establish a genuinely quantum internal structure. A degenerate label may remain effectively classical if all physically distinguished observables acting on it are simultaneously diagonalizable.

We now show that this is not the case for the natural relational observables associated with pairwise neutral comparison. Different ways of asking which constituents form rotational singlets give incompatible observables, even though every such observable commutes with the total rotational action. In the four-carrier neutral sector these observables generate the complete qubit matrix algebra. More generally, the algebra generated by pairwise singlet relations coincides with the commutant of the collective $SU(2)$ action on $(\mathbb{C}^2)^{\otimes N}$.

For two minimal projective carriers i and j , define the pairwise singlet projector

$$P_{ij}^{(0)} := \frac{1}{4} (I - \sigma_i \cdot \sigma_j), \quad (62)$$

where

$$\sigma_i \cdot \sigma_j = \sum_{a=1}^3 \sigma_i^a \sigma_j^a. \quad (63)$$

Equivalently,

$$P_{ij}^{(0)} = |s_{ij}\rangle\langle s_{ij}|,$$

where

$$|s_{ij}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle).$$

The operator $P_{ij}^{(0)}$ asks a purely relational question: whether the two selected projective carriers combine into the neutral rotational channel. It does not assign an absolute spin direction to either constituent.

Because the two-particle singlet is rotationally invariant,

$$[P_{ij}^{(0)}, J_{\text{tot}}^a] = 0 \quad a = 1, 2, 3. \quad (64)$$

Hence

$$P_{ij}^{(0)} \in (\mathcal{A}_N^{\text{coll}})'. \quad (65)$$

Every pairwise neutral-comparison projector is therefore invisible to the collective rotational representation in the sense that it preserves each total-spin sector.

The important point is that different relational questions need not be compatible.

Consider first the four-carrier invariant sector

$$\mathcal{I}_4 \simeq \mathbb{C}^2$$

introduced in Section 2. We use the logical basis

$$\{|0_L\rangle, |1_L\rangle\},$$

where

$$|0_L\rangle = |s_{12}\rangle |s_{34}\rangle$$

and

$$|1_L\rangle = \frac{1}{\sqrt{3}} (|t_{12}^+ t_{34}^-\rangle - |t_{12}^0 t_{34}^0\rangle + |t_{12}^- t_{34}^+\rangle).$$

In this basis,

$$P_{12}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (66)$$

A recoupling calculation gives

$$P_{23}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (67)$$

The sign of the off-diagonal entries depends on the overall phase convention for $|1_L\rangle$, but all algebraic conclusions below are phase independent.

The two observables do not commute:

$$[P_{12}^{(0)}, P_{23}^{(0)}] = \frac{\sqrt{3}}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -\frac{i\sqrt{3}}{4} \sigma_y. \quad (68)$$

This gives the first explicit relational incompatibility.

Proposition 3.1 (Incompatible neutral relations). *On the four-carrier neutral sector I_4 , the two relational questions*

“are carriers 1, 2 neutral?”

and

“are carriers 2, 3 neutral?”

are represented by noncommuting projectors:

$$[P_{12}^{(0)}, P_{23}^{(0)}] \neq 0. \quad (69)$$

Both observables commute with collective rotation.

The proposition separates two notions that would coincide in a classical relational model. Each projector is a scalar under collective geometry, but the two scalar relations cannot be assigned simultaneous sharp values in general. The noncommutativity resides in the relational recoupling structure rather than in an absolute geometric direction.

The result is stronger than the existence of one nonzero commutator. The two projectors generate all operators on the relational qubit.

Introduce logical Pauli operators

$$X_L, \quad Y_L, \quad Z_L$$

in the basis

$$\{|0_L\rangle, |1_L\rangle\}.$$

Equation (66) gives

$$P_{12}^{(0)} = \frac{1}{2} (I + Z_L). \quad (70)$$

Equation (67) becomes

$$P_{23}^{(0)} = \frac{1}{2} I - \frac{\sqrt{3}}{4} X_L - \frac{1}{4} Z_L. \quad (71)$$

Hence

$$Z_L = 2P_{12}^{(0)} - I, \quad (72)$$

and

$$X_L = -\frac{4}{\sqrt{3}} \left(P_{23}^{(0)} - \frac{1}{2} I + \frac{1}{4} Z_L \right). \quad (73)$$

The commutator then supplies

$$Y_L = \frac{4i}{\sqrt{3}} [P_{12}^{(0)}, P_{23}^{(0)}]. \quad (74)$$

Theorem 3.2 (Relational qubit algebra). *The unital complex algebra generated on I_4 by $P_{12}^{(0)}$ and $P_{23}^{(0)}$ is the full matrix algebra:*

$$\text{Alg} \{P_{12}^{(0)}, P_{23}^{(0)}\} = \text{End}(I_4) \simeq M_2(\mathbb{C}). \quad (75)$$

Proof. Equation (72) shows that the generated algebra contains Z_L . Equation (73) then shows that it contains X_L . Their commutator contains Y_L . Therefore the algebra contains

$$I, \quad X_L, \quad Y_L, \quad Z_L,$$

which form a basis of $M_2(\mathbb{C})$. □

Thus a complete qubit algebra is generated by observables that are individually rotational scalars. No additional microscopic two-state label is required once the relational multiplicity sector is present.

The four-carrier example is the smallest geometrically neutral realization, but the underlying algebraic statement is general. Let

$$\mathcal{R}_N := \text{Alg} \{P_{ij}^{(0)}; 1 \leq i < j \leq N\} \subseteq \text{End}(\mathcal{H}_N) \quad (76)$$

be the algebra generated by all pairwise neutral-comparison projectors.

Let

$$S_{ij}$$

denote the unitary operator that exchanges carriers i and j . On two spin- $\frac{1}{2}$ carriers, exchange acts as $+1$ on the triplet and -1 on the singlet. Therefore

$$P_{ij}^{(0)} = \frac{1}{2} (I - S_{ij}), \quad (77)$$

or equivalently

$$S_{ij} = I - 2P_{ij}^{(0)}. \quad (78)$$

The permutation action commutes with collective $SU(2)$:

$$[S_{ij}, U_N(g)] = 0. \quad (79)$$

Conversely, Schur–Weyl duality states that the commutant of the collective action on

$$(\mathbb{C}^2)^{\otimes N}$$

is the image of the complex group algebra of the permutation group S_N . Since transpositions generate S_N , and since every transposition is related to a singlet projector by Eq. (78), the pairwise relational projectors generate the complete collective commutant.

Theorem 3.3 (Relational generation of the collective commutant). *For N minimal projective comparison carriers,*

$$\mathcal{R}_N = (\mathcal{A}_N^{\text{coll}})'. \quad (80)$$

Under the multiplicity decomposition

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)},$$

this algebra acts as

$$\mathcal{R}_N \simeq \bigoplus_j I_{V_j} \otimes \text{End}(M_j^{(N)}). \quad (81)$$

In particular, after restriction to any fixed total-spin sector, pairwise neutral relations generate the complete matrix algebra on the associated multiplicity space.

Proof. By Eq. (64),

$$\mathcal{R}_N \subseteq (\mathcal{A}_N^{\text{coll}})'.$$

Equation (78) shows that $\mathcal{R}_N$ contains every transposition operator S_{ij} , and hence contains the image of the group algebra $\mathbb{C}[S_N]$.

Schur–Weyl duality identifies this image with the full commutant of the collective $SU(2)$ action. Therefore

$$\mathcal{R}_N = (\mathcal{A}_N^{\text{coll}})'.$$

The block form follows from Theorem 2.4. □

The theorem substantially sharpens the interpretation of relational multiplicity. The commutant is not merely an abstract location in which internal observables might be placed. For composites of minimal projective carriers, it is generated by explicit pairwise relational observables.

A local generating set can be chosen using only nearest-neighbor relations. Define

$$e_i := 2P_{i,i+1}^{(0)}, \quad i = 1, \dots, N-1. \quad (82)$$

These operators satisfy

$$e_i^2 = 2e_i, \quad (83)$$

$$e_i e_{i\pm 1} e_i = e_i, \quad (84)$$

$$[e_i, e_j] = 0, \quad |i - j| > 1. \quad (85)$$

Thus the relational commutant may equivalently be represented as the appropriate finite-dimensional spin- $\frac{1}{2}$ image of the Temperley–Lieb algebra with loop parameter 2.

This formulation is useful because the noncommutativity has a direct relational origin. Distant neutral-pair questions commute, whereas overlapping pair questions are constrained by Eq. (84) and are generally incompatible. The algebraic structure therefore records how different relational decompositions overlap.

The preceding results concern the operator algebra. We now examine where the corresponding information is stored.

Let

$$\mathcal{I}_N = \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes N} \right]$$

for even N , and let

$$\rho$$

be any density operator whose support lies entirely in $\mathcal{I}_N$. Since the collective action is trivial on $\mathcal{I}_N$,

$$U_N(g) \rho U_N(g)^\dagger = \rho \quad \text{for all } g \in SU(2). \quad (86)$$

Let

$$\rho_i = \text{Tr}_{\bar{i}} \rho$$

denote the reduced state of carrier i , where the trace is taken over all other carriers. Partial tracing Eq. (86) gives

$$u_{\frac{1}{2}}(g) \rho_i u_{\frac{1}{2}}(g)^\dagger = \rho_i \quad \text{for all } g \in SU(2). \quad (87)$$

By irreducibility of the fundamental representation, Schur's lemma implies that ρ_i is proportional to the identity.

Theorem 3.4 (One-carrier geometric blindness). *For every normalized state ρ supported on a total-singlet sector $\mathcal{I}_N$,*

$$\rho_i = \frac{1}{2} I \quad (88)$$

for every carrier i . Consequently,

$$\text{Tr}(\rho J_i^a) = 0 \quad (89)$$

for all i and a .

The theorem shows that no one-carrier state contains information about the logical state encoded in a neutral relational sector. Distinct logical states may be perfectly distinguishable by relational observables while every individual carrier has the same maximally mixed reduced state.

There is a stronger operator statement. Let

$$P_{\mathcal{I}_N}$$

denote the orthogonal projector onto $\mathcal{I}_N$.

Theorem 3.5 (Vanishing of one-body vector operators). *For every carrier i ,*

$$P_{\mathcal{I}_N} J_i^a P_{\mathcal{I}_N} = 0, \quad a = 1, 2, 3. \quad (90)$$

More generally, the restriction to $\mathcal{I}_N$ of any operator transforming as a nontrivial irreducible tensor under collective $SU(2)$ vanishes between singlet states.

Proof. Let

$$|\Phi\rangle, |\Psi\rangle \in \mathcal{I}_N$$

and define

$$v^a = \langle \Phi | J_i^a | \Psi \rangle.$$

Both states are invariant under collective rotation. Since J_i^a transforms as a vector,

$$U_N(g) J_i^a U_N(g)^{-1} = R(g)^a_b J_i^b,$$

where $R(g)$ is the spin-one rotation matrix. Therefore

$$v^a = R(g)^a_b v^b$$

for every rotation g . The only vector invariant under all rotations is the zero vector. Hence

$$v^a = 0.$$

Since this holds for arbitrary $|\Phi\rangle, |\Psi\rangle \in \mathcal{I}_N$, Eq. (90) follows. The general statement follows by the same argument for any nontrivial irreducible tensor representation. $\square$

The theorem gives a useful protection principle. Consider a perturbation linear in one-carrier geometric vectors,

$$V = \sum_{i,a} h_{ia} J_i^a. \quad (91)$$

Then

$$P_{\mathcal{I}_N} V P_{\mathcal{I}_N} = 0. \quad (92)$$

Thus such a perturbation produces no first-order splitting within the neutral multiplicity space.

Equation (92) does not imply complete immunity from the perturbation. The operator V may couple the singlet sector to non-singlet states and therefore cause leakage if those states are dynamically accessible. If the neutral sector is separated from its complement by a gap and the non-singlet sector can be integrated out perturbatively, the leading internal effect is second order.

Let

$$P = P_{\mathcal{I}_N}, \quad Q = I - P,$$

and suppose the unperturbed Hamiltonian has energy E_0 on the protected sector. The standard second-order effective operator has the form

$$H_{\text{eff}}^{(2)} = -PVQ(H_Q - E_0)^{-1}QVP, \quad (93)$$

where

$$H_Q = QH_0Q.$$

Products of two vector operators contain rotational scalar components, so Eq. (93) naturally generates relational operators such as

$$\mathbf{J}_i \cdot \mathbf{J}_j. \quad (94)$$

Equivalently, these may be expressed in terms of $P_{ij}^{(0)}$.

This gives a simple hierarchy:

$$\text{one-body geometric vector} \longrightarrow \text{no first-order internal action,}$$

whereas

$$\text{second-order comparison} \longrightarrow \text{relational scalar dynamics.}$$

The conclusion depends on the existence of a protected low-energy singlet sector and should not be interpreted as a general statement that all geometric perturbations decouple from relational information.

We next quantify how much quantum information can be stored while retaining exact collective neutrality.

For

$$N = 2n,$$

Section 2 gave

$$\dim \mathcal{I}_{2n} = C_n = \frac{1}{n+1} \binom{2n}{n}, \quad (95)$$

where C_n is the n -th Catalan number. Its large- n asymptotic form is

$$C_n \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}. \quad (96)$$

Hence

$$\log C_n = 2n \log 2 - \frac{3}{2} \log n - \frac{1}{2} \log \pi + o(1). \quad (97)$$

Writing

$$N = 2n,$$

one obtains

$$\log_2 \dim \mathcal{I}_N = N - \frac{3}{2} \log_2 N + O(1). \quad (98)$$

Therefore

$$\lim_{\substack{N \rightarrow \infty \\ N \text{ even}}} \frac{\log_2 \dim \mathcal{I}_N}{N} = 1. \quad (99)$$

Theorem 3.6 (Asymptotically extensive neutral relational capacity). *The total-singlet sector of $N = 2n$ minimal projective comparison carriers can encode*

$$k_N = \log_2 \dim \mathcal{I}_N = N - \frac{3}{2} \log_2 N + O(1) \quad (100)$$

logical qubits of Hilbert-space capacity. The asymptotic information rate satisfies

$$\lim_{N \rightarrow \infty} \frac{k_N}{N} = 1$$

along even N .

The terminology “logical qubits of Hilbert-space capacity” is intentional. If $\dim \mathcal{I}_N$ is not an exact power of two, the quantity

$$\log_2 \dim \mathcal{I}_N$$

is an information-capacity measure rather than the number of exactly factorized qubit subsystems. The theorem states that imposing exact collective rotational neutrality costs only a subextensive amount of Hilbert-space capacity.

This result has a direct structural interpretation. A large composite comparison system can satisfy

$$\mathbf{J}_{\text{tot}} = 0$$

exactly while retaining almost the full exponential complexity of the underlying tensor-product Hilbert space. The absence of a collective geometric spin therefore does not imply microscopic simplicity.

The combination of the preceding theorems gives three increasingly strong statements.

First, multiplicity spaces exist whenever the same collective irreducible representation occurs more than once:

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}.$$

Second, pairwise relational observables generate the entire collective commutant:

$$\mathcal{R}_N = \left(\mathcal{A}_N^{\text{coll}} \right)'.$$

Third, in the exactly neutral sector,

$$\mathcal{I}_{2n} = M_0^{(2n)},$$

all one-carrier geometric information is erased while the relational information capacity remains asymptotically extensive.

These facts motivate the interpretation of multiplicity as an internal relational quantum structure. The word “internal” must nevertheless retain its restricted meaning. The results established so far concern invisibility under the collective rotational comparison action. They do not prove invisibility under every local gravitational observable, and they do not yet construct a spacetime gauge symmetry.

There is also no contradiction with the fact that the underlying projective carriers possess spin. The logical degrees of freedom are not new constituent spins. They describe inequivalent coupling patterns among the same spinorial comparison carriers after the collective representation has been fixed.

Nor does the equality

$$\mathcal{R}_N = \left(\mathcal{A}_N^{\text{coll}} \right)'$$

imply that the full unitary group of every multiplicity block is dynamically realized. It establishes the available relational observable algebra at one composite system. A physical internal comparison symmetry must additionally specify which relational transformations can be transported consistently between different spacetime events.

This is the point at which the problem becomes geometric again. Suppose that a protected multiplicity sector

$$M_x$$

is selected at every event x , but its embedding in a larger comparison Hilbert space varies with position. Local relational bases then form a vector bundle, and comparison between such bases introduces a connection. A mere change of basis inside a fixed multiplicity subspace produces only pure gauge, whereas genuine motion of the physical subspace produces curvature.

The next section develops this transition:

relational multiplicity $\longrightarrow$ multiplicity bundle $\longrightarrow$ Grassmannian curvature $\longrightarrow$ comparison holonomy.

It will also establish the distinction between the maximal kinematical frame group $U(m)$ of an m -dimensional multiplicity space and the smaller group actually generated by relational holonomy.

4 Multiplicity Bundles, Grassmannian Curvature, and Holonomy

The preceding sections were formulated primarily at a fixed event. Once a collective geometric representation has been selected, its multiplicity space supports a noncommutative relational algebra that is invisible to collective rotational comparison. A gauge-like structure, however, cannot arise from a single multiplicity space considered in isolation. It requires a rule for comparing relational states at distinct spacetime events.

We therefore assume that a protected multiplicity sector can be selected smoothly over a spacetime region. The word “protected” will be used in a kinematical sense in the present section: the selected subspace is assumed to remain separated well enough from its complement that parallel comparison within the subspace is meaningful. The microscopic mechanism responsible for such protection, whether spectral, topological, energetic, or otherwise, is not specified here.

Let

$$\mathcal{R} \longrightarrow M \tag{101}$$

be a complex Hermitian vector bundle of finite rank N , representing an ambient local comparison space. Let

$$D : \Gamma(\mathcal{R}) \longrightarrow \Omega^1(M, \mathcal{R}) \tag{102}$$

be a Hermitian connection with curvature

$$F_{\mathcal{R}} = D^2. \tag{103}$$

The ambient bundle need not itself be interpreted as a physical internal gauge bundle. It is a book-keeping space in which the physically selected multiplicity sector is embedded.

Suppose that, at every x in an open region of M , an m -dimensional protected relational subspace

$$M_x \subset \mathcal{R}_x \tag{104}$$

is specified smoothly. The family of subspaces defines a rank- m Hermitian subbundle

$$\mathcal{M} \longrightarrow M. \quad (105)$$

Locally choose an orthonormal frame of $\mathcal{M}$,

$$Z = (|m_1\rangle, \dots, |m_m\rangle), \quad Z^\dagger Z = I_m. \quad (106)$$

Here Z may be viewed fiberwise as an isometric map

$$Z : \mathbb{C}^m \longrightarrow \mathcal{R}_x.$$

The orthogonal projector onto the physical multiplicity space is

$$P = ZZ^\dagger, \quad P^2 = P, \quad P^\dagger = P. \quad (107)$$

The frame Z is not unique. For every smooth

$$U : M \longrightarrow U(m),$$

the transformation

$$Z \longmapsto ZU \quad (108)$$

leaves P unchanged. Thus $U(m)$ appears immediately as the local frame freedom of an m -dimensional complex multiplicity bundle.

This statement alone has no dynamical content. It expresses the freedom to choose coordinates in the same protected subspace. The distinction between frame freedom and physical comparison curvature will be central below.

The ambient connection induces a connection on $\mathcal{M}$ by orthogonal projection:

$$\nabla^{\mathcal{M}} := PD|_{\Gamma(\mathcal{M})}. \quad (109)$$

In the local frame Z , its connection one-form is

$$A := Z^\dagger DZ. \quad (110)$$

Since D preserves the Hermitian metric,

$$A^\dagger = -A,$$

so

$$A \in \Omega^1(M, \mathfrak{u}(m)).$$

Under the frame change Eq. (108),

$$A \longmapsto A^U = U^{-1}AU + U^{-1}dU. \quad (111)$$

Thus A transforms exactly as a $U(m)$ connection.

Proposition 4.1 (Induced relational connection). *Let $\mathcal{M} \subset \mathcal{R}$ be a smooth rank- m Hermitian subbundle and let D be a Hermitian connection on $\mathcal{R}$. Then*

$$\nabla^{\mathcal{M}} = PD$$

defines a Hermitian connection on $\mathcal{M}$. In any local orthonormal frame Z , its connection matrix is

$$A = Z^\dagger DZ$$

and transforms according to Eq. (111).

Proof. If

$$\psi = Zc$$

is a section of $\mathcal{M}$, then

$$PD\psi = ZZ^\dagger D(Zc).$$

Using

$$Z^\dagger Z = I_m$$

gives

$$PD(Zc) = Z\left(dc + Z^\dagger DZ c\right).$$

Hence the induced covariant derivative is represented by

$$d + A, \quad A = Z^\dagger DZ.$$

For $Z' = ZU$,

$$\begin{aligned} A' &= Z'^\dagger DZ' \\ &= U^\dagger Z^\dagger D(ZU) \\ &= U^{-1}AU + U^{-1}dU. \end{aligned} \tag{112}$$

□

It is useful to decompose the ambient derivative of the frame into a component tangent to the protected subspace and a component normal to it. Define

$$K := (I - P)DZ. \tag{113}$$

Then

$$DZ = ZA + K. \tag{114}$$

The matrix-valued one-form K measures how the physical multiplicity subspace moves through the ambient comparison space. It is the analogue of a second fundamental form for the embedded vector bundle.

Under

$$Z \mapsto ZU,$$

one has

$$K \mapsto KU. \tag{115}$$

Hence quantities such as

$$K^\dagger \wedge K$$

transform covariantly in the multiplicity fiber.

The curvature of the induced connection is

$$F_A := dA + A \wedge A. \tag{116}$$

A direct calculation gives the corresponding Gauss-type relation.

Theorem 4.2 (Relational Gauss curvature formula). *Let*

$$A = Z^\dagger DZ, \quad K = (I - P)DZ.$$

Then the curvature of the protected multiplicity bundle satisfies

$$F_A = Z^\dagger F_{\mathcal{R}} Z + K^\dagger \wedge K. \quad (117)$$

Equivalently, as an endomorphism of the subbundle,

$$F_{\mathcal{M}} = PF_{\mathcal{R}}P + P(DP \wedge DP)P. \quad (118)$$

Proof. Using

$$A = Z^\dagger DZ$$

and the metric compatibility of D , one obtains

$$dA = (DZ)^\dagger \wedge DZ + Z^\dagger F_{\mathcal{R}} Z. \quad (119)$$

From

$$DZ = ZA + K$$

and

$$Z^\dagger K = 0, \quad A^\dagger = -A,$$

it follows that

$$(DZ)^\dagger \wedge DZ = -A \wedge A + K^\dagger \wedge K. \quad (120)$$

Substitution into Eq. (119) gives Eq. (117).

For the projector form, note that

$$PDPP = 0$$

as a consequence of differentiating $P^2 = P$, and that the off-diagonal part of DP is precisely the motion of the selected subspace relative to its complement. Restricting

$$P(DP \wedge DP)P$$

to the frame Z yields

$$K^\dagger \wedge K.$$

This gives Eq. (118). □

The theorem separates two possible sources of internal comparison curvature. The first,

$$Z^\dagger F_{\mathcal{R}} Z,$$

is inherited from the ambient comparison bundle. The second,

$$K^\dagger \wedge K,$$

is generated by the changing embedding of the protected multiplicity sector itself.

The latter is particularly important when the ambient comparison space is taken to be locally flat:

$$F_{\mathcal{R}} = 0. \quad (121)$$

Then

$$F_A = K^\dagger \wedge K = Z^\dagger [P(DP \wedge DP)P] Z. \quad (122)$$

Thus nontrivial internal curvature can arise even when the larger comparison space has no curvature of its own.

This mechanism must be distinguished sharply from a change of basis inside a fixed multiplicity space.

Proposition 4.3 (No curvature from basis motion alone). *Suppose the ambient comparison bundle is flat on a simply connected local patch and choose a trivialization in which*

$$D = d.$$

If the physical multiplicity projector is constant,

$$P(x) = P_0, \quad (123)$$

then every orthonormal frame of the protected subspace may be written locally as

$$Z(x) = Z_0 U(x), \quad U(x) \in U(m), \quad (124)$$

and the induced connection is pure gauge:

$$A = U^{-1} dU. \quad (125)$$

Consequently,

$$F_A = 0. \quad (126)$$

Proof. If $P = P_0$, the image of every $Z(x)$ is the same m -dimensional subspace. Any two orthonormal frames of that subspace differ by a unitary matrix, so

$$Z = Z_0 U.$$

With $dZ_0 = 0$,

$$A = U^{-1} dU.$$

The Maurer–Cartan identity gives

$$dA + A \wedge A = 0.$$

□

The proposition gives a simple no-go statement:

$$\text{local basis freedom} \not\Rightarrow \text{physical internal curvature}. \quad (127)$$

The formal appearance of a gauge transformation law is therefore insufficient to establish an emergent gauge field.

In a flat ambient bundle, nonzero curvature requires genuine motion of the physical subspace:

$$F_A \neq 0 \implies DP \neq 0. \quad (128)$$

The converse need not hold. A nonconstant projector can move in such a way that

$$P(DP \wedge DP)P$$

vanishes on a particular region or family of tangent two-planes. Thus $DP \neq 0$ is a necessary kinematical condition for subspace-generated curvature, not a sufficient one.

The projector description makes the Grassmannian structure explicit. A rank- m orthogonal projector in $\mathbb{C}^N$ defines an m -plane, and therefore a point of the complex Grassmannian

$$\text{Gr}(m, N). \quad (129)$$

Accordingly, a varying protected multiplicity sector determines a map

$$P : M \longrightarrow \text{Gr}(m, N). \quad (130)$$

The bundle $\mathcal{M}$ is the pullback of the tautological rank- m bundle over $\text{Gr}(m, N)$, while Eq. (118) is the pullback of its universal connection curvature.

Proposition 4.4 (Grassmannian realization of relational geometry). *Let the ambient comparison bundle be locally trivial of rank N , and let $P(x)$ be a smooth rank- m orthogonal projector. Then:*

1. *P defines a smooth map*

$$M \longrightarrow \text{Gr}(m, N);$$

2. *the protected multiplicity bundle is the pullback of the tautological bundle;*
3. *the induced comparison connection is the pullback of the universal Grassmannian connection;*
4. *in a flat ambient trivialization its curvature is*

$$F_{\mathcal{M}} = P(dP \wedge dP)P.$$

The Grassmannian formulation separates the intrinsic multiplicity dimension m from the number N of ambient comparison directions required to realize its motion. A fixed m -dimensional subspace carries only frame freedom. Nontrivial comparison geometry appears because the physical m -plane itself explores

$$\text{Gr}(m, N).$$

In the stable limit,

$$\text{Gr}(m, \infty) \simeq BU(m), \quad (131)$$

up to homotopy. Thus sufficiently large ambient comparison spaces have the standard topological capacity to realize general rank- m complex vector bundles. This fact is useful for classifying possible relational bundles, but it should not be interpreted dynamically. It does not imply that MS comparison dynamics realizes every element of $BU(m)$, nor that every allowed bundle supports a physical gauge field.

The same embedding data also define a matrix-valued quantum geometric tensor. In local coordinates x^μ , write

$$K_\mu = (I - P)D_\mu Z. \quad (132)$$

Define

$$Q_{\mu\nu} := K_\mu^\dagger K_\nu. \quad (133)$$

Its Hermitian symmetric part measures the local change of the protected subspace, while its antisymmetric part determines the induced curvature:

$$F_{\mu\nu}^{\text{ext}} = K_{\mu}^{\dagger} K_{\nu} - K_{\nu}^{\dagger} K_{\mu}. \quad (134)$$

When the ambient connection is flat,

$$F_{\mu\nu} = F_{\mu\nu}^{\text{ext}}.$$

Thus the same local data that measure how rapidly the relational subspace moves also control its comparison curvature.

A simple norm estimate follows from the Cauchy–Schwarz inequality. Defining

$$g_{\mu\mu} := \text{tr}(K_{\mu}^{\dagger} K_{\mu}), \quad (135)$$

one obtains

$$\|F_{\mu\nu}^{\text{ext}}\|_{\text{HS}} \leq 2\sqrt{g_{\mu\mu}g_{\nu\nu}}, \quad (136)$$

where $\|\cdot\|_{\text{HS}}$ denotes the Hilbert–Schmidt norm. Consequently, large relational curvature requires correspondingly nontrivial variation of the protected subspace.

The curvature is a local object. The physically realized internal comparison group is more naturally associated with its holonomy.

Fix a base point $x \in M$. For every closed path

$$\gamma : x \longrightarrow x,$$

parallel transport by ∇^M defines a unitary operator

$$U_{\gamma}^M \in U(M_x). \quad (137)$$

Define the relational holonomy group

$$G_{\text{hol}}(x) := \overline{\text{Hol}_x(\nabla^M)} \subseteq U(M_x). \quad (138)$$

The closure is included to emphasize that the physically relevant compact group generated by all comparison loops is being considered.

This gives a sharper notion of internal comparison symmetry than the full frame group

$$U(m).$$

The latter is the maximal kinematical freedom in choosing a local multiplicity basis. The former is the subgroup actually generated by transport around comparison loops.

Definition 4.5 (Dynamical internal comparison group). *For a protected relational multiplicity bundle $\mathcal{M} \rightarrow M$ equipped with its induced comparison connection, define the dynamical internal comparison group at x by*

$$G_{\text{int}}^{\text{dyn}}(x) := G_{\text{hol}}(x). \quad (139)$$

It is understood up to conjugation under a change of multiplicity frame.

Thus

$$G_{\text{int}}^{\text{dyn}} \subseteq G^{\text{kin}} = U(m). \quad (140)$$

Equality is possible, but it is not automatic.

The Lie algebra of the holonomy group is determined by curvature. In the standard smooth finite-dimensional setting, the Ambrose–Singer theorem gives the relevant characterization [?, ?].

Theorem 4.6 (Curvature generation of relational holonomy). *Let ∇^M be the relational connection on a connected region. The restricted holonomy Lie algebra at a base point x is generated by curvature operators transported back to x :*

$$\mathfrak{g}_{\text{hol}}(x) = \text{Lie} \left\langle U_\gamma^{-1} F_y(X, Y) U_\gamma \right\rangle, \quad (141)$$

where y ranges over points connected to x , γ is a path from x to y , and

$$X, Y \in T_y M.$$

This theorem provides a direct operational criterion for non-Abelian internal comparison. A multiplicity space of dimension greater than one is not sufficient. The curvature operators must generate a noncommutative Lie algebra under parallel transport and Lie closure.

The two-dimensional case illustrates the distinction particularly clearly. Suppose

$$\dim M_x = 2.$$

Any anti-Hermitian curvature matrix may be decomposed as

$$F = \frac{1}{2} \text{tr}(F) I + F^{(0)}, \quad F^{(0)} \in \mathfrak{su}(2). \quad (142)$$

If all transported traceless curvature operators are proportional to one fixed generator,

$$F^{(0)} \in \mathbb{R} X \quad \text{for one fixed } X \in \mathfrak{su}(2), \quad (143)$$

their Lie closure is Abelian. By contrast, two noncommuting traceless curvature directions generate the full $\mathfrak{su}(2)$.

Proposition 4.7 (Two-dimensional non-Abelian holonomy criterion). *Let M have complex rank two.*

If every parallel-transported traceless curvature operator lies in one one-dimensional subalgebra of $\mathfrak{su}(2)$, then

$$\mathfrak{g}_{\text{hol}}^{(0)} \subseteq \mathfrak{u}(1). \quad (144)$$

If there exist two transported traceless curvature operators $X, Y \in \mathfrak{su}(2)$ satisfying

$$[X, Y] \neq 0, \quad (145)$$

then their Lie closure is

$$\text{Lie}\langle X, Y \rangle = \mathfrak{su}(2). \quad (146)$$

If, in addition, an independent nonzero trace curvature is present, the full holonomy algebra may extend to

$$\mathfrak{u}(2).$$

Proof. The first statement follows because a one-dimensional Lie subalgebra is Abelian. For the second, identify

$$\mathfrak{su}(2) \simeq \mathbb{R}^3$$

with Lie bracket proportional to the vector cross product. Two nonproportional elements have a nonzero commutator linearly independent of both. The three elements therefore span all of $\mathfrak{su}(2)$.

The identity component commutes with the traceless algebra, so an independent central curvature direction enlarges $\mathfrak{su}(2)$ to $\mathfrak{u}(2)$. $\square$

This example makes clear why multiplicity dimension should not be confused with gauge-group identification. A relational qubit provides the kinematical envelope

$$U(2),$$

but the dynamically generated comparison group may be trivial, $U(1)$, $SU(2)$, $U(2)$, or a smaller discrete extension, depending on the actual connection and its curvature.

The same distinction applies in higher rank. The decomposition

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}$$

determines the maximal blockwise commutant

$$\prod_j U(M_j^{(N)}),$$

but no finite- N representation-theoretic argument determines the holonomy subgroup realized by physical comparison dynamics.

There is a further reason not to identify one such holonomy group immediately with a universal internal gauge group. Different composite sectors may carry different multiplicity bundles:

$$\mathcal{M}_r, \quad \mathcal{M}_s, \quad \mathcal{M}_t, \dots$$

and hence apparently different connections

$$A^{(r)}, \quad A^{(s)}, \quad A^{(t)}.$$

Without an additional compatibility requirement, these could describe a collection of unrelated Berry or Wilczek–Zee geometries rather than one common internal symmetry.

A physical interaction theory requires more. If relational sectors can be combined, their transport must respect the maps by which they combine. Comparison along a path and fusion of sectors should not depend on the order in which the two operations are performed.

This observation marks the transition from multiplicity geometry to internal symmetry reconstruction. The structures established in the present section may be summarized as

$$\begin{aligned} \text{protected relational multiplicity} &\longrightarrow \mathcal{M} \longrightarrow M, \\ \text{moving physical subspace} &\longrightarrow P : M \longrightarrow \text{Gr}(m, N), \\ \text{projected comparison} &\longrightarrow A = Z^\dagger DZ, \\ \text{subspace twisting} &\longrightarrow F_A, \\ \text{curvature Lie closure} &\longrightarrow G_{\text{int}}^{\text{dyn}} \subseteq U(m). \end{aligned} \tag{147}$$

The first three arrows are geometric once a protected multiplicity subbundle has been selected. The fourth distinguishes genuine subspace geometry from mere frame choice. The fifth identifies the group actually generated by comparison loops rather than the maximal group permitted by the multiplicity dimension.

The next section imposes the missing inter-sector condition. If localized relational sectors possess fusion maps

$$C_{rs}^t : M_r \otimes M_s \longrightarrow M_t,$$

then consistency requires fusion and inter-event transport to commute. This condition will lead from separate multiplicity holonomies to a tensor-compatible common comparison group and will provide the appropriate setting for charge addition, conjugate sectors, and a Tannaka-type internal group reconstruction.

5 Fusion-Compatible Transport and Internal Group Reconstruction

The preceding section associates a connection and a holonomy group with each protected multiplicity bundle. Taken separately, however, such bundles need not define a common internal symmetry. Different relational sectors could carry unrelated Wilczek–Zee connections,

$$A^{(r)}, \quad A^{(s)}, \quad A^{(t)},$$

with no principle requiring them to be representations of the same underlying comparison structure.

A genuine internal gauge organization requires an additional consistency condition. If localized relational sectors can be combined, then inter-event comparison must preserve the maps by which they combine. In other words, transporting two sectors and then fusing them must give the same result as fusing them first and transporting the composite.

Let

$$r, s, t, \dots$$

denote localized relational sector labels, and let

$$M_r(x), \quad M_s(x), \quad M_t(x)$$

be their multiplicity fibers at an event x . Suppose that an admissible fusion channel

$$r \otimes s \longrightarrow t$$

is represented by a linear map

$$C_{rs}^t(x) : M_r(x) \otimes M_s(x) \longrightarrow M_t(x). \tag{148}$$

The map need not be invertible and may possess a nontrivial multiplicity of its own. When several independent fusion maps exist, an additional intertwiner label may be included; we suppress it initially for clarity.

Let

$$\gamma : x \longrightarrow y$$

be a comparison path. Denote the parallel transports in the corresponding multiplicity bundles by

$$U_{\gamma}^{(r)} : M_r(x) \longrightarrow M_r(y), \quad (149)$$

and similarly for s and t .

The basic compatibility condition is the commutativity of the diagram

$$\begin{array}{ccc} M_r(x) \otimes M_s(x) & \xrightarrow{C_{rs}^t(x)} & M_t(x) \\ \downarrow U_{\gamma}^{(r)} \otimes U_{\gamma}^{(s)} & & \downarrow U_{\gamma}^{(t)} \\ M_r(y) \otimes M_s(y) & \xrightarrow{C_{rs}^t(y)} & M_t(y). \end{array}$$

Equivalently,

$$U_{\gamma}^{(t)} C_{rs}^t(x) = C_{rs}^t(y) (U_{\gamma}^{(r)} \otimes U_{\gamma}^{(s)}). \quad (150)$$

Definition 5.1 (Fusion-compatible comparison transport). *A family of inter-event transports*

$$\{U_{\gamma}^{(r)}\}_r$$

is called fusion compatible if Eq. (150) holds for every admissible fusion map and every comparison path for which the sectors are defined.

This condition is stronger than requiring each

$$U_{\gamma}^{(r)}$$

to be unitary. It correlates the comparison transformations of different sectors and prevents them from being chosen independently.

Infinitesimally, fusion compatibility imposes a relation among the corresponding connection one-forms. Let

$$\nabla^{(r)} = d + A^{(r)}$$

denote the local connection on M_r , and similarly for s and t . The tensor-product connection on

$$M_r \otimes M_s$$

is

$$\nabla^{(r \otimes s)} = \nabla^{(r)} \otimes I + I \otimes \nabla^{(s)}. \quad (151)$$

Fusion compatibility implies that

$$C_{rs}^t$$

is covariantly constant with respect to these connections.

Proposition 5.2 (Infinitesimal fusion compatibility). *Equation (150) for all sufficiently short paths is equivalent locally to*

$$\nabla C_{rs}^t = 0, \quad (152)$$

where

$$\nabla C_{rs}^t = dC_{rs}^t + A^{(t)} C_{rs}^t - C_{rs}^t (A^{(r)} \otimes I + I \otimes A^{(s)}). \quad (153)$$

Proof. Let

$$\gamma_\epsilon$$

be an infinitesimal path with tangent vector X . Expanding the transports to first order gives

$$U_{\gamma_\epsilon}^{(r)} = I - \epsilon A_X^{(r)} + O(\epsilon^2),$$

with analogous expressions for s and t , while

$$C_{rs}^t(y) = C_{rs}^t(x) + \epsilon d_X C_{rs}^t + O(\epsilon^2).$$

Inserting these expansions into Eq. (150) and comparing the first-order terms yields

$$d_X C_{rs}^t + A_X^{(t)} C_{rs}^t - C_{rs}^t (A_X^{(r)} \otimes I + I \otimes A_X^{(s)}) = 0.$$

Since X is arbitrary, this is Eq. (152). □

In a local choice of fusion frame for which

$$dC_{rs}^t = 0,$$

the condition reduces to

$$A_X^{(t)} C_{rs}^t = C_{rs}^t (A_X^{(r)} \otimes I + I \otimes A_X^{(s)}). \quad (154)$$

Thus the connections seen by different sectors cannot be specified independently once nontrivial fusion channels are present.

The same statement holds at the level of curvature. Applying a second covariant derivative to

$$\nabla C_{rs}^t = 0$$

gives

$$F^{(t)} C_{rs}^t = C_{rs}^t (F^{(r)} \otimes I + I \otimes F^{(s)}). \quad (155)$$

Hence curvature itself must respect the fusion network.

Corollary 5.3 (Fusion-compatible curvature). *If C_{rs}^t is covariantly constant, then the curvature operators in the three sectors satisfy Eq. (155). In particular, the holonomy Lie algebras acting on different sectors are constrained to be representation-compatible.*

This observation already shows why a collection of unrelated multiplicity connections is insufficient. The internal comparison geometry must preserve the full algebra of composition maps among localized sectors.

It is useful to formulate this requirement categorically.

Let

$$\mathcal{C}_{\text{rel}}$$

denote a category whose objects are localized relational sectors and whose morphisms are admissible local intertwiners. We assume that the physical composition of sectors supplies a tensor product

$$\otimes : \mathcal{C}_{\text{rel}} \times \mathcal{C}_{\text{rel}} \longrightarrow \mathcal{C}_{\text{rel}},$$

together with a tensor unit

$$\mathbf{1}.$$

Direct sums, subobjects, and conjugate sectors will be introduced when required below.

At each event x , define a fiber assignment

$$\omega_x : C_{\text{rel}} \longrightarrow \text{Hilb}_{\text{fd}}, \quad r \longmapsto M_r(x). \quad (156)$$

For a morphism

$$f : r \rightarrow s,$$

the map

$$\omega_x(f) : M_r(x) \rightarrow M_s(x)$$

is its action on the corresponding multiplicity fibers.

The tensor structure requires coherent identifications

$$\omega_x(r \otimes s) \simeq \omega_x(r) \otimes \omega_x(s), \quad (157)$$

together with

$$\omega_x(\mathbf{1}) \simeq \mathbb{C}.$$

Thus ω_x is a candidate unitary tensor fiber functor.

The internal symmetry at a fixed event is then not naturally the full product of the individual unitary groups

$$\prod_r U(M_r(x)).$$

Only those families of unitaries that preserve every relational morphism and every tensor product are admissible.

Definition 5.4 (Tensor-compatible internal automorphism). *A tensor-compatible automorphism of ω_x is a family*

$$g = \{g_r\}_r, \quad g_r \in U(M_r(x)), \quad (158)$$

satisfying:

$$g_s \omega_x(f) = \omega_x(f) g_r \quad \text{for every } f : r \rightarrow s, \quad (159)$$

$$g_{r \otimes s} = g_r \otimes g_s, \quad (160)$$

$$g_{\mathbf{1}} = I. \quad (161)$$

The group of such automorphisms is denoted

$$G_x := \text{Aut}^\otimes(\omega_x). \quad (162)$$

The naturality condition is the abstract form of preserving all admissible relational observables and fusion intertwiners. The monoidal condition says that the transformation of a composite is determined by the transformations of its constituents.

Accordingly,

$$G_x \subseteq \prod_r U(M_r(x)), \quad (163)$$

typically as a much smaller subgroup.

This is the appropriate point at which an internal symmetry group may be reconstructed. The reconstruction, however, requires assumptions stronger than those established by the finite multiplicity analysis alone.

Assumption 5.5 (Tensor reconstruction hypotheses). *The localized relational sectors form a rigid unitary tensor category C_{rel} with simple tensor unit, finite-dimensional morphism spaces, direct sums and subobjects, and appropriate conjugates. Where a symmetric reconstruction is invoked, a compatible symmetric structure is additionally assumed. The fiber functor*

$$\omega_x : C_{\text{rel}} \rightarrow \text{Hilb}_{\text{fd}}$$

is assumed faithful, unitary, and tensor compatible.

Under standard compact Tannaka–Krein or Doplicher–Roberts hypotheses, these data reconstruct a compact group from tensor automorphisms of the fiber functor [?, ?].

Theorem 5.6 (Conditional internal group reconstruction). *Under Assumption 5.5, define*

$$G_x = \text{Aut}^{\otimes}(\omega_x).$$

Then G_x is a compact group acting consistently on every multiplicity fiber,

$$\rho_r : G_x \longrightarrow U(M_r(x)), \quad (164)$$

and every relational intertwiner is G_x -equivariant. Under the appropriate reconstruction hypotheses, the relational tensor category is realized as a tensor category of finite-dimensional representations of the reconstructed compact group.

The theorem is conditional in two distinct senses. First, the abstract Tannaka–Krein and Doplicher–Roberts reconstruction results are standard mathematics rather than consequences specific to MS. Second, the present paper has not derived all hypotheses of Assumption 5.5 from the original operational response. The MS-specific claim is instead that projective comparison carriers and their multiplicity spaces supply a concrete candidate source for the objects and fibers entering such a reconstruction.

The significance of the theorem is nevertheless substantial. A universal internal group is no longer guessed from one multiplicity dimension such as

$$m = 2 \quad \text{or} \quad m = 3.$$

It is reconstructed from the full compatibility structure of all relational sectors and all admissible fusion maps.

The same logic extends from a single event to spacetime comparison. For a path

$$\gamma : x \rightarrow y,$$

suppose that the family of sector transports

$$U_{\gamma}^{(r)}$$

is natural with respect to all morphisms and monoidal with respect to tensor products. Then it defines a tensor-natural unitary isomorphism

$$U_{\gamma} : \omega_x \Rightarrow \sim \omega_y. \quad (165)$$

Explicitly,

$$U_{\gamma}^{(s)} \omega_x(f) = \omega_y(f) U_{\gamma}^{(r)}, \quad (166)$$

$$U_{\gamma}^{(r \otimes s)} = U_{\gamma}^{(r)} \otimes U_{\gamma}^{(s)}, \quad (167)$$

$$U_{\gamma}^{(1)} = I. \quad (168)$$

Fusion compatibility Eq. (150) is a special case of Eq. (166).
Choose a reference event

$$x_0$$

and write

$$\omega_0 := \omega_{x_0}.$$

At every event x , consider the set

$$\mathcal{P}_x := \text{Iso}^\otimes(\omega_0, \omega_x) \quad (169)$$

of unitary tensor-natural isomorphisms from the reference fiber functor to the fiber functor at x .

The group

$$G := \text{Aut}^\otimes(\omega_0)$$

acts on the right by composition:

$$p \cdot g := p \circ g. \quad (170)$$

Whenever $\mathcal{P}_x$ is nonempty, this action is free and transitive.

Proposition 5.7 (Tensor-comparison torsor). *For each x , if*

$$\mathcal{P}_x = \text{Iso}^\otimes(\omega_0, \omega_x)$$

is nonempty, then $\mathcal{P}_x$ is a right torsor for

$$G = \text{Aut}^\otimes(\omega_0).$$

Proof. If

$$p, q \in \mathcal{P}_x,$$

then

$$p^{-1}q \in \text{Aut}^\otimes(\omega_0) = G,$$

and hence

$$q = p \cdot (p^{-1}q).$$

This proves transitivity. If

$$p \cdot g = p,$$

then composition with p^{-1} gives

$$g = I,$$

so the action is free. □

If the torsors $\mathcal{P}_x$ vary smoothly with x , they assemble into a principal bundle

$$\mathcal{P} \longrightarrow M \quad (171)$$

with structure group G .

This gives a more precise reconstruction of a common internal bundle than the independent Grassmannian description of the previous section. The Grassmannian construction explains how one multiplicity sector acquires a connection. The tensor-functor construction explains when the connections seen by all sectors are representations of one principal comparison connection.

Theorem 5.8 (Conditional comparison-principal-bundle reconstruction). *Assume that:*

1. *the fiber functors ω_x satisfy Assumption 5.5;*
2. *the sets*

$$\text{Iso}^{\otimes}(\omega_0, \omega_x)$$

are nonempty and vary smoothly with x ;

3. *inter-event comparison defines smooth tensor-natural parallel transport satisfying composition under concatenation of paths.*

Then the family

$$\mathcal{P}_x = \text{Iso}^{\otimes}(\omega_0, \omega_x)$$

defines a principal G -bundle

$$\mathcal{P} \rightarrow M, \quad G = \text{Aut}^{\otimes}(\omega_0),$$

and the comparison transport defines a principal connection on $\mathcal{P}$. For every relational sector r ,

$$M_r \simeq \mathcal{P} \times_{\rho_r} M_r(x_0),$$

and its induced connection is

$$A^{(r)} = \rho_{r*}(A), \tag{172}$$

where A is the common principal connection.

The theorem identifies a concrete criterion for gauge universality. The connections

$$A^{(r)}$$

appearing in different matter-like sectors should not be viewed as unrelated matrix-valued one-forms. They must arise as representation images of one principal comparison connection.

The logical distinction is

$$\begin{aligned} \text{one multiplicity bundle} &\longrightarrow \text{a local } U(m) \text{ frame freedom,} \\ \text{one multiplicity connection} &\longrightarrow \text{one holonomy subgroup,} \\ \text{fusion-compatible family of sectors} &\longrightarrow \text{one common tensor-compatible internal group.} \end{aligned} \tag{173}$$

Only the final line provides the structure expected of a universal internal gauge principle.

The fusion law also gives immediate selection rules in Abelian sectors. Let the reconstructed group contain a factor $U(1)$, and suppose sector r transforms as

$$\rho_r(e^{i\theta}) = e^{iq_r\theta}. \tag{174}$$

Here the normalization of q_r depends on the chosen generator of the $U(1)$ factor.

Suppose

$$C_{rs}^t \neq 0$$

is a fusion intertwiner. Equivariance requires

$$\rho_t(g)C_{rs}^t = C_{rs}^t (\rho_r(g) \otimes \rho_s(g)). \tag{175}$$

For

$$g = e^{i\theta},$$

this gives

$$e^{iq_t\theta}C_{rs}^t = e^{i(q_r+q_s)\theta}C_{rs}^t.$$

Since the intertwiner is nonzero and this must hold for every θ ,

$$q_t = q_r + q_s. \quad (176)$$

Corollary 5.9 (Abelian charge additivity). *For a $U(1)$ factor reconstructed from tensor-compatible relational comparison, every nonzero fusion channel*

$$r \otimes s \rightarrow t$$

satisfies

$$q_t = q_r + q_s.$$

If a conjugate sector satisfies

$$r \otimes \bar{r} \rightarrow \mathbf{1},$$

then

$$q_{\bar{r}} = -q_r, \quad (177)$$

while the tensor unit has

$$q_1 = 0.$$

Thus charge conservation need not be introduced as an independent arithmetic rule once a $U(1)$ comparison factor and fusion-compatible transport are present. It is the Abelian form of tensor compatibility.

The same reasoning extends to non-Abelian groups. If

$$C_{rs}^t \neq 0,$$

then

$$\rho_t$$

must occur in the representation-theoretic decomposition of

$$\rho_r \otimes \rho_s.$$

Fusion selection rules are therefore ordinary intertwiner constraints of the reconstructed internal group.

This observation should not be reversed without care. A desired collection of phenomenological fusion rules does not automatically prove that the underlying relational category satisfies the hypotheses of a compact-group reconstruction. Non-group-like tensor categories, braided categories, or more general quantum symmetries are mathematically possible. The emergence of an ordinary compact gauge group therefore requires the corresponding tensor and symmetry assumptions rather than following from multiplicity alone.

There is also an important distinction between the reconstructed internal group

$$G = \text{Aut}^\otimes(\omega_0)$$

and the holonomy group of a particular connection. The former determines the allowed common internal frame transformations of the complete relational category. The latter is the subgroup actually generated by a chosen comparison connection:

$$G_{\text{hol}} \subseteq G. \quad (178)$$

A theory may therefore possess a kinematically reconstructed group G while a specific vacuum or background connection realizes a smaller holonomy subgroup.

Conversely, curvature in several representations cannot be chosen independently if it originates from one principal connection. If

$$F \in \Omega^2(M, \text{Ad } \mathcal{P})$$

is the principal curvature, then

$$F^{(r)} = \rho_{r*}(F). \quad (179)$$

Equation (155) is then automatic because the fusion maps are intertwiners.

This gives a useful consistency test for a microscopic MS comparison model. Suppose an effective calculation produces sector-dependent Berry connections

$$A_{\text{eff}}^{(r)}.$$

To interpret them as manifestations of one internal gauge structure, it is not enough that each transforms as a connection. They must satisfy at least:

$$\nabla C_{rs}^t = 0 \quad (180)$$

for all physical fusion channels, and their holonomies must admit a common tensor-compatible lift.

Failure of these conditions would indicate that the apparent gauge connections are separate geometric phases rather than representations of one universal internal comparison field.

A particularly transparent diagnostic is provided by closed comparison loops. For

$$\gamma : x \rightarrow x,$$

fusion compatibility requires

$$U_{\gamma}^{(t)} C_{rs}^t = C_{rs}^t (U_{\gamma}^{(r)} \otimes U_{\gamma}^{(s)}). \quad (181)$$

Thus the holonomy matrices across all sectors form a tensor-compatible family. At the reference event they define an element of

$$\text{Aut}^{\otimes}(\omega_0).$$

Consequently,

$$\text{Hol}_{x_0} \subseteq \text{Aut}^{\otimes}(\omega_0). \quad (182)$$

This relation unifies the two constructions developed so far. The Grassmannian description determines how a protected multiplicity sector may acquire curvature. The tensor-compatible reconstruction determines which curvatures in different sectors may belong to one common internal geometry.

The resulting hierarchy is

$$\begin{aligned} &\text{projective carriers} \longrightarrow \text{relational multiplicities,} \\ &\text{relational multiplicities} \longrightarrow \text{noncommutative commutant algebra,} \\ &\text{varying protected multiplicities} \longrightarrow \text{sector connections and holonomies,} \\ &\text{fusion-compatible comparison} \longrightarrow \text{Aut}^{\otimes}(\omega), \\ &\text{smooth tensor-natural transport} \longrightarrow \text{common principal comparison connection.} \end{aligned} \quad (183)$$

The construction does not yet establish Yang–Mills dynamics. A principal bundle and a connection may exist kinematically even if the connection is entirely slaved to slower microscopic variables and possesses no independent transverse excitations. Likewise, nontrivial holonomy does not imply a massless gauge boson.

Nor does the reconstruction determine a particular phenomenological gauge group. The group

$$\text{Aut}^\otimes(\omega)$$

depends on the actual relational sectors, their fusion rules, their intertwiners, and the category realized by the microscopic comparison dynamics. Nothing established so far singles out

$$SU(3) \times SU(2) \times U(1),$$

its quotient by a central subgroup, the observed matter representations, or their charge assignments.

The result is instead structural. Internal gauge universality can be formulated without postulating one independent gauge connection for every matter sector. If localized relational sectors possess a common fusion-compatible comparison structure, then their multiplicity fibers can be organized by one tensor automorphism group, and their individual connections become representation images of one principal connection.

This provides a sharper meaning of “internal” in the present MS construction. Internal degrees of freedom are not merely operators that commute with one chosen geometric representation. A stable internal gauge structure requires three levels of consistency:

- commutant compatibility : internal observables preserve geometric representation,
- transport compatibility : internal states can be compared between events, (184)
- fusion compatibility : comparison preserves composition of sectors.

Only after the third level does a common internal group become naturally reconstructible.

The remaining question is dynamical. The connection reconstructed above may still be a composite Berry connection determined algebraically by a family of protected subspaces. To obtain a gauge field in the field-theoretic sense, one must show that suitable comparison modes survive as independent long-wavelength degrees of freedom and acquire a finite kinetic stiffness without an infrared mass term.

The next section therefore separates

internal gauge structure

from

propagating gauge-field dynamics.

It formulates a minimal infrared criterion for the liberation of a comparison connection, identifies the role of the transverse two-point kernel, and clarifies which conclusions would and would not justify the interpretation of an emergent Yang–Mills sector.

6 From Comparison Connections to Gauge-Field Dynamics

The preceding sections reconstruct an increasingly strong form of internal comparison structure. Composite projective sectors generate geometry-invisible multiplicity spaces; varying protected multiplicities define vector bundles with induced connections; curvature and holonomy select the transformations actually generated by comparison transport; and fusion-compatible transport can organize the separate sector connections as representations of one principal internal connection.

None of these statements, however, implies that the reconstructed connection is a propagating gauge field.

This distinction is essential. A Berry or Wilczek–Zee connection may be completely determined by the motion of a protected subspace,

$$A = Z^\dagger DZ, \quad (185)$$

with

$$P = ZZ^\dagger.$$

If P is itself a functional of more fundamental slow variables

$$\lambda^A(x),$$

then the spacetime connection may take the composite form

$$A_\mu = A_A(\lambda) \partial_\mu \lambda^A. \quad (186)$$

Such a connection can possess nonzero curvature and nontrivial holonomy while carrying no independent local degree of freedom.

We shall refer to this situation as a slaved comparison connection.

Definition 6.1 (Slaved comparison connection). *A comparison connection A is called slaved on a given effective description if its local value is completely determined, up to frame gauge, by other dynamical fields and their derivatives,*

$$A = \mathcal{A}[\lambda], \quad (187)$$

so that A cannot be varied independently while the remaining effective variables are held fixed.

The definition is independent of whether the connection has curvature. Indeed,

$$F[A] \neq 0$$

does not imply the existence of an independent vector excitation. Curvature measures the failure of parallel comparison to be path independent; it does not count phase-space degrees of freedom.

This observation gives a first dynamical no-go statement.

Proposition 6.2 (Holonomy is not sufficient for gauge propagation). *The existence of a nontrivial multiplicity bundle, a connection A , nonzero curvature F_A , or nontrivial holonomy*

$$G_{\text{hol}} \neq \{I\}$$

does not by itself imply the existence of an independently propagating gauge field.

Proof. All four structures may occur for a family of protected eigenspaces parametrized by other fields λ , with the connection determined algebraically or differentially by

$$A = \mathcal{A}[\lambda].$$

The physical phase space then contains the independent variables λ , but no additional freely variable connection coordinate. Parallel transport may nevertheless have nontrivial curvature and holonomy. Hence neither property is sufficient to establish an independent gauge-field mode. $\square$

To identify a propagating gauge sector one must therefore study the effective dynamics of fluctuations of the comparison connection.

Suppose that the fusion-compatible construction of Section 5 has reconstructed a compact internal group G and a principal comparison connection A . Let

$$\bar{A}$$

denote a reference comparison configuration and write

$$A = \bar{A} + a, \quad (188)$$

where a is an adjoint-valued one-form.

A necessary first step is that a survive as an independent variable of the low-energy effective theory rather than being eliminated algebraically in terms of the remaining fields.

Let

$$\Gamma_{\text{eff}}[A, \Phi]$$

denote the one-particle-irreducible effective action after all degrees of freedom above a specified infrared scale have been integrated out, with Φ denoting the remaining non-connection fields. We say that a connection fluctuation is dynamically independent if the effective field space contains variations

$$\delta A \neq 0, \quad \delta \Phi = 0,$$

that are not pure gauge and are not removed by algebraic constraint equations.

This requirement separates a true low-energy connection coordinate from a mere rewriting of the projector variables.

Around a translationally invariant vacuum, or locally in a regime where a momentum-space approximation is meaningful, the quadratic effective action for small fluctuations may be written schematically as

$$\Gamma_{\text{eff}}^{(2)} = \frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} a_\mu^A(-k) \Gamma_{AB}^{\mu\nu}(k) a_\nu^B(k). \quad (189)$$

Here A, B are adjoint indices of the reconstructed internal group.

For a Lorentz-invariant unbroken sector, the transverse part can be written as

$$\Gamma_{AB}^{\mu\nu}(k) = \Pi_{AB}^T(k^2) P_T^{\mu\nu}(k) + \Pi_{AB}^L(k) P_L^{\mu\nu}(k), \quad (190)$$

where, away from $k^2 = 0$,

$$P_T^{\mu\nu}(k) = \eta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}, \quad P_L^{\mu\nu}(k) = \frac{k^\mu k^\nu}{k^2}. \quad (191)$$

The longitudinal part depends on the gauge implementation and gauge fixing. The physical infrared information relevant to a massless vector excitation is contained in the transverse kernel.

When a local derivative expansion exists,

$$\Pi_{AB}^T(k^2) = (M_A^2)_{AB} + K_{AB} k^2 + O(k^4). \quad (192)$$

The matrix

$$M_A^2$$

is an infrared vector mass matrix, while

$$K_{AB}$$

is the leading kinetic stiffness.

For a simple irreducible internal factor with an invariant generator normalization, symmetry typically reduces

$$K_{AB}$$

to a scalar multiple of the invariant quadratic form. We then write

$$K_{AB} = K_A \delta_{AB}. \quad (193)$$

The corresponding transverse propagator behaves near a massless pole as

$$D_{\mu\nu}^{AB}(k) \sim \frac{i(K^{-1})^{AB}}{k^2 + i0} P_{\mu\nu}^T(k), \quad (194)$$

up to gauge-dependent longitudinal terms and conventional normalization.

This motivates the following criterion.

Definition 6.3 (Connection liberation criterion). *A reconstructed internal comparison connection is said to be liberated as a massless infrared gauge field on a specified sector if the following conditions hold:*

1. *its low-energy fluctuation a_μ is an independent effective field rather than a slaved functional of other variables;*
2. *the effective action possesses the local G -comparison redundancy, or an equivalent set of Ward or BRST identities, required to remove unphysical polarizations;*
3. *the unbroken transverse sector satisfies*

$$M_A^2 = 0; \quad (195)$$

4. *the leading transverse kinetic form is finite and positive in the physical normalization,*

$$0 < K_A < \infty. \quad (196)$$

The four requirements have different logical roles. The first asserts the existence of an independent mode. The second identifies that mode as a gauge connection rather than a generic massless vector. The third gives the long-range pole, and the fourth gives a finite healthy response coefficient.

A vanishing quadratic mass coefficient alone is therefore insufficient.

Proposition 6.4 (Masslessness alone is not connection liberation). *The condition*

$$M_A^2 = 0$$

in a quadratic effective kernel does not by itself establish an emergent gauge field.

Proof. A massless quadratic vector mode may still fail to possess a local gauge redundancy, may be constrained by additional degrees of freedom, may acquire inconsistent interactions, or may be a composite response whose apparent vector coordinate is not independently variable. The gauge-compatible effective identities and the independence condition in Definition 6.3 are therefore logically additional. $\square$

Conversely, under the liberation conditions the infrared response contains a massless transverse pole.

Proposition 6.5 (Long-range response of a liberated comparison connection). *Suppose a comparison connection satisfies Definition 6.3 and couples linearly to a conserved internal current,*

$$S_{\text{int}} = \int d^4x a_\mu^A J_A^\mu, \quad \partial_\mu J_A^\mu = 0 \quad (197)$$

at the linearized level. Then its leading long-wavelength transverse response is

$$a_\mu^A(k) \sim (K^{-1})^{AB} \frac{P_{\mu\nu}^T(k)}{k^2 + i0} J_B^\nu(k), \quad (198)$$

up to normalization and higher-derivative corrections.

The pole distinguishes a liberated connection from a purely local algebraic response. A slaved connection may generate geometric phases, but it need not mediate an independent long-range interaction.

Once the normalization of the reconstructed group generators and the connection in the covariant derivative has been fixed, the kinetic stiffness may be related to the usual gauge coupling convention. Writing

$$S_{\text{YM}} = -\frac{1}{4g^2} \int d^4x F_{\mu\nu}^A F_A^{\mu\nu} + \dots, \quad (199)$$

one identifies schematically

$$K_A \propto g^{-2}. \quad (200)$$

The proportionality depends on generator normalization and on whether the coupling is placed in the definition of the connection or in the matter covariant derivative.

Accordingly, the physical statement is more invariantly expressed as

$$\text{large gauge stiffness} \iff \text{small low-energy gauge response}, \quad (201)$$

once conventions have been fixed.

This parallels the response interpretation used elsewhere in the MS reconstruction program: a coupling constant is associated with the susceptibility of a comparison sector rather than with the existence of the underlying geometric structure itself.

The origin of the kinetic stiffness remains dynamical. One possible mechanism is that microscopic relational degrees of freedom generate an effective action for the comparison connection when they are integrated out. Schematically, let χ denote such microscopic modes and consider

$$\exp(i\Gamma_{\text{ind}}[A]) = \int \mathcal{D}\chi \exp(iS[\chi, A]). \quad (202)$$

If the eliminated sector is gapped, locality and symmetry may permit a derivative expansion of the form

$$\Gamma_{\text{ind}}[A] = \int d^4x \left[-\frac{K_A}{4} F_{\mu\nu}^A F_A^{\mu\nu} + \frac{c_1}{\Delta^2} (D_\rho F_{\mu\nu})^2 + \dots \right], \quad (203)$$

where Δ is a microscopic gap scale.

Equation (203) is a possible infrared mechanism, not a result established by the multiplicity construction. The sign and magnitude of K_A , the absence of a mass term, and the stability of the putative gauge phase must all be derived from a microscopic comparison dynamics.

If the modes being integrated out are gapless, the effective action need not admit a purely local derivative expansion. Nonanalytic structures such as

$$F_{\mu\nu} \log\left(\frac{-D^2}{\mu^2}\right) F^{\mu\nu} \quad (204)$$

may occur. A single constant K_A should therefore be interpreted only within a regime where a local leading-order description is meaningful.

For a non-Abelian sector, the quadratic kernel is not the complete test. The same group structure reconstructed from fusion must govern the nonlinear vertices. A Yang–Mills-like infrared action takes the form

$$F = dA + A \wedge A, \quad (205)$$

so the quadratic, cubic, and quartic interaction terms are not independent. They must be related by the same local comparison symmetry.

Thus a microscopic theory that happens to generate several massless vector modes with the correct two-point functions would still be insufficient if their higher-point vertices failed the corresponding gauge identities.

We may summarize the non-Abelian requirement schematically as

$$\begin{aligned} \text{two-point sector} &\longrightarrow \text{massless transverse pole,} \\ \text{three- and four-point sectors} &\longrightarrow \text{same } G\text{-covariant interaction algebra,} \\ \text{all orders} &\longrightarrow \text{Ward/Slavnov–Taylor consistency.} \end{aligned} \quad (206)$$

This leads to a useful distinction between three levels of gauge emergence:

$$\begin{aligned} \text{bundle level :} & \quad \text{multiplicity fibers admit local frame transformations,} \\ \text{connection level :} & \quad \text{inter-event comparison has curvature and holonomy,} \\ \text{field level :} & \quad \text{the connection has independent gauge-consistent dynamics.} \end{aligned} \quad (207)$$

The present paper establishes a route through the first two levels and formulates explicit criteria for the third.

There is also a possible failure mode in which the connection remains massive. If

$$M_A^2 > 0,$$

the corresponding response is short ranged. This does not invalidate the multiplicity or holonomy construction. It means only that the selected comparison vacuum does not realize an unbroken long-range gauge sector. Depending on the microscopic theory, such a mass could reflect explicit breaking, spontaneous symmetry breaking, confinement-related effective behavior, or the absence of a genuine gauge redundancy. The present kinematical construction does not distinguish these possibilities.

A second possible failure occurs when

$$K_A = 0.$$

Then the leading local kinetic term is absent, and higher-derivative or nonlocal terms control the comparison dynamics. A third occurs when the effective kinetic form has an unhealthy sign, indicating an instability or that the assumed low-energy variables are not appropriate physical modes.

These failure conditions are useful because they make the proposed gauge emergence program falsifiable at the level of an explicit microscopic model. A candidate MS comparison dynamics must do more than produce a degenerate multiplicity space. It must generate a protected subspace with non-trivial transport, satisfy fusion compatibility across sectors, and yield an infrared effective connection with the correct masslessness, stiffness, and gauge identities.

The complete chain developed in this paper can now be stated as

$$\begin{aligned}
&\text{minimal projective comparison carriers} \longrightarrow \text{relational multiplicity,} \\
&\text{overlapping neutral relations} \longrightarrow \text{noncommutative internal algebra,} \\
&\text{protected subspace variation} \longrightarrow \text{connection, curvature, and holonomy,} \\
&\text{fusion-compatible transport} \longrightarrow \text{common internal principal structure,} \\
&\text{connection liberation} \longrightarrow \text{candidate propagating gauge sector.}
\end{aligned} \tag{208}$$

Only the first four lines are developed structurally in the present paper. The final line is formulated as a dynamical target. A microscopic realization must determine whether that target is actually achieved.

7 Discussion and Open Problems

The purpose of this paper has been to investigate whether internal quantum structure can arise from composites of projective comparison sectors without being inserted as an independent microscopic label. The analysis begins after the quantum-soldering and relative-comparison constructions: the existence of localized minimal projective carriers is treated as an input, while the consequences of composing such carriers are developed explicitly.

The first result is representation theoretic but has a direct relational interpretation. For

$$N$$

minimal projective carriers,

$$\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N},$$

collective rotational comparison gives

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}. \tag{209}$$

The factor V_j carries the geometry-visible total spin, while $M_j^{(N)}$ records the multiplicity with which the same total geometric representation occurs.

The multiplicity is

$$m_j^{(N)} = \binom{N}{\frac{N}{2} - j} - \binom{N}{\frac{N}{2} - j - 1}. \tag{210}$$

Already for three carriers, the total spin- $\frac{1}{2}$ sector possesses a two-dimensional multiplicity. For four carriers, the total singlet sector is itself two-dimensional:

$$\mathcal{I}_4 = \text{Inv}_{SU(2)} \left[(\mathbb{C}^2)^{\otimes 4} \right] \simeq \mathbb{C}^2. \tag{211}$$

Thus exact collective geometric neutrality does not imply absence of internal quantum information.

The second result concerns the observable algebra acting on that information. The natural pairwise neutral-comparison operators

$$P_{ij}^{(0)}$$

commute with collective rotation but not, in general, with one another. On the four-carrier invariant sector,

$$P_{12}^{(0)}$$

and

$$P_{23}^{(0)}$$

generate

$$M_2(\mathbb{C}).$$

More generally, the algebra generated by all pairwise singlet projectors is the complete commutant of the collective $SU(2)$ action:

$$\text{Alg} \{P_{ij}^{(0)}\} = (\mathcal{A}_N^{\text{coll}})'. \quad (212)$$

Under Eq. (209), this becomes

$$(\mathcal{A}_N^{\text{coll}})' \simeq \bigoplus_j I_{V_j} \otimes \text{End}(M_j^{(N)}). \quad (213)$$

The internal algebra is therefore not introduced by assigning an additional quantum number to each constituent. It is generated by incompatible questions about how the same projective carriers are coupled while the total geometric representation is held fixed.

The use of Schur–Weyl duality and the associated Temperley–Lieb representation in establishing this result is standard representation theory. The structural point specific to the present program is the physical placement of this commutant: the elementary carriers arise from projective comparison, while the multiplicity algebra is interpreted as relational information invisible to collective geometry.

The third result is a strong form of one-body geometric blindness. Every state supported on a total-singlet sector satisfies

$$\rho_i = \frac{1}{2}I \quad (214)$$

for every constituent, and

$$P_{I_N} J_i^a P_{I_N} = 0. \quad (215)$$

Consequently, a perturbation linear in one-body vector operators does not split the neutral multiplicity sector at first order. Relational scalar operators arise naturally at higher order.

This blindness does not reduce the neutral state space to negligible size. For

$$N = 2n,$$

$$\dim \mathcal{I}_{2n} = \frac{1}{n+1} \binom{2n}{n}, \quad (216)$$

so

$$\log_2 \dim \mathcal{I}_N = N - \frac{3}{2} \log_2 N + O(1). \quad (217)$$

The information capacity of the exactly neutral sector is therefore asymptotically extensive.

These results motivate, but do not by themselves establish, the language of internal degrees of freedom. Throughout the paper, “internal” first means invisible to the collective rotational representation. A complete gravitational theory may contain additional local geometric observables that resolve structures not seen by this restricted collective action. Extending the commutant statement from the rotational comparison algebra to a fully localized gravitational observable algebra remains an important problem.

The fourth result concerns inter-event comparison. A protected multiplicity space selected smoothly over spacetime defines a Hermitian vector bundle

$$\mathcal{M} \rightarrow M.$$

Embedding it in an ambient comparison bundle and choosing a local orthonormal frame Z gives

$$P = ZZ^\dagger, \quad A = Z^\dagger DZ. \quad (218)$$

The induced curvature satisfies the Gauss-type relation

$$F_A = Z^\dagger F_{\mathcal{R}} Z + K^\dagger \wedge K, \quad K = (I - P)DZ. \quad (219)$$

In a flat ambient bundle this reduces to

$$F_{\mathcal{M}} = P(dP \wedge dP)P. \quad (220)$$

This immediately distinguishes genuine subspace geometry from basis gauge freedom. If

$$P(x) = P_0$$

is constant and only the basis changes,

$$Z(x) = Z_0 U(x),$$

then

$$A = U^{-1} dU$$

and

$$F_A = 0.$$

A multiplicity dimension m therefore provides a maximal frame group $U(m)$, but it does not generate physical curvature by itself.

The physical comparison group associated with one selected multiplicity bundle is more sharply characterized by holonomy:

$$G_{\text{int}}^{\text{dyn}} = \overline{\text{Hol}(\nabla^{\mathcal{M}})} \subseteq U(m). \quad (221)$$

Its connected Lie algebra is generated by parallel-transported curvature operators. In particular, a rank-two multiplicity bundle has the kinematical frame group $U(2)$, but its actual holonomy may remain Abelian unless the curvature explores noncommuting directions.

This distinction prevents a common overinterpretation:

$$\dim M = m \quad \not\Rightarrow \quad G_{\text{internal}} = U(m). \quad (222)$$

The fifth result addresses universality across sectors. Separate multiplicity bundles could carry unrelated Berry connections. To obtain one common internal comparison structure, transport must preserve fusion.

For a fusion map

$$C_{rs}^t : M_r \otimes M_s \rightarrow M_t,$$

the required pathwise compatibility is

$$U_\gamma^{(t)} C_{rs}^t(x) = C_{rs}^t(y) \left(U_\gamma^{(r)} \otimes U_\gamma^{(s)} \right). \quad (223)$$

Infinitesimally,

$$\nabla C_{rs}^t = 0. \quad (224)$$

Curvature then satisfies the corresponding intertwining relation.

Under the additional hypotheses required for a compact tensor reconstruction, the localized relational sectors define a fiber functor

$$\omega_x : C_{\text{rel}} \rightarrow \text{Hilb}_{\text{fd}},$$

and the common internal group is

$$G_x = \text{Aut}^\otimes(\omega_x). \quad (225)$$

If tensor-natural inter-event comparisons vary smoothly, the torsors

$$\text{Iso}^\otimes(\omega_0, \omega_x)$$

assemble into a principal bundle, and the connections in the individual relational sectors become representation images of one principal connection:

$$A^{(r)} = \rho_{r*}(A). \quad (226)$$

This reconstruction should not be confused with a derivation of a new Tannaka–Krein theorem. The categorical reconstruction results themselves are standard. The question specific to MS is whether its projective comparison sectors dynamically provide the localized objects, fusion maps, conjugates, and transport structure required by those theorems.

An Abelian factor gives a particularly transparent consequence. If

$$r \otimes s \rightarrow t$$

is an allowed fusion channel and

$$\rho_r(e^{i\theta}) = e^{iq_r\theta},$$

then tensor compatibility gives

$$q_t = q_r + q_s. \quad (227)$$

Thus additive charge conservation is the Abelian representation of fusion-compatible comparison rather than an independent arithmetic rule.

This does not determine the numerical values of the charges. Their normalization and the spectrum of admissible representations depend on the actual reconstructed group and sector category.

The sixth result is primarily a boundary statement. Bundle geometry, curvature, holonomy, and even a common principal connection are not sufficient to establish propagating Yang–Mills dynamics. A comparison connection may remain entirely slaved to other microscopic variables.

For a genuine massless gauge-field regime, the present paper formulates the connection-liberation conditions. The connection fluctuation must survive as an independent low-energy variable; the effective theory must possess the appropriate local comparison redundancy or equivalent gauge identities; the transverse mass matrix must vanish; and the leading kinetic stiffness must be finite and positive.

The quadratic transverse kernel then has the form

$$\Pi_T(k^2) = K_A k^2 + O(k^4), \quad (228)$$

and the propagator contains the long-range pole

$$D_T(k) \sim \frac{1}{K_A} \frac{1}{k^2 + i0}. \quad (229)$$

Once normalization conventions are fixed,

$$K_A \propto g^{-2}. \quad (230)$$

The gauge coupling is therefore interpreted as a response stiffness of the liberated internal comparison sector.

For a non-Abelian theory this two-point condition is only the beginning. Cubic, quartic, and higher vertices must be controlled by the same reconstructed group and satisfy the corresponding gauge identities. No such microscopic derivation is supplied here.

The complete logical chain obtained in the paper is therefore

$$\begin{aligned} \Xi = -I + \text{minimality} &\longrightarrow V_{\frac{1}{2}}, \\ V_{\frac{1}{2}}^{\otimes N} &\longrightarrow V_j \otimes M_j^{(N)}, \\ \text{pairwise neutral comparison} &\longrightarrow \text{End}(M_j^{(N)}), \\ \text{protected spatial variation} &\longrightarrow \mathcal{M}, A, F, G_{\text{hol}}, \\ \text{fusion-compatible transport} &\longrightarrow \text{Aut}^{\otimes}(\omega), \\ \text{dynamical liberation} &\longrightarrow \text{candidate gauge-field sector}. \end{aligned} \quad (231)$$

The first arrow remains conditional because

$$\Xi = -I$$

selects all half-integer projective spins, not uniquely $j = \frac{1}{2}$. The last arrow remains dynamical because neither representation multiplicity nor Berry geometry forces an independent massless connection.

Several problems must therefore be solved before this construction can be identified with observed particle gauge structure.

The most immediate is the microscopic origin of the minimal projective carrier. A relative-comparison model can conditionally generate spin- $\frac{1}{2}$ projective endpoints, but localization, deconfinement, and a controlled continuum limit must be demonstrated rather than assumed.

A second problem is protection of the multiplicity sector. The present bundle construction assumes that a relational subspace can be selected smoothly across spacetime. A concrete theory must identify the spectral gap, constraint, topological protection, or other mechanism that prevents uncontrolled mixing with the complementary comparison Hilbert space.

A third problem is locality. The finite tensor products used here identify the algebra of relational multiplicity at one composite system. A field theory requires localized sector algebras, causal composition, transport between bounded regions, and a consistent relation between these structures and the operationally reconstructed spacetime geometry.

A fourth problem is the derivation of the fusion category itself. Fusion rules, associators, conjugates, and possible symmetry or braiding structures cannot be chosen solely for the purpose of reproducing a desired gauge group. They must follow from the comparison dynamics. In particular, a generic tensor category need not reconstruct an ordinary compact group; more general quantum or braided symmetry structures are possible unless additional hypotheses are established.

A fifth problem is group selection. Nothing in the present analysis derives

$$SU(3) \times SU(2) \times U(1),$$

a quotient thereof, or any other specific phenomenological internal group. The dimensions

$$2, 3, \dots$$

that occur as finite multiplicities must not be identified with observed gauge-group dimensions. The physical group is determined by the full tensor-compatible sector structure, not by one multiplicity block.

A sixth problem is chirality. The projective invariant detects half-integer rotational spin parity but does not select one Weyl chirality, and the internal multiplicity construction does not by itself distinguish left- and right-handed relativistic representations. Chiral gauge structure therefore requires further input.

A seventh problem is spin–statistics. The elementary projective carriers used throughout the paper need not be fermions solely because

$$\Xi = -I.$$

A relativistic spin–statistics identification requires locality, positive-energy conditions, covariance, and the corresponding field-theoretic hypotheses.

An eighth problem is anomaly consistency. Even if a chiral internal group and matter representation content are reconstructed, the quantum comparison theory must still be consistent under the corresponding local symmetry. Anomaly cancellation would then become a necessary condition for the quantum implementation of comparison equivalence.

A ninth problem is connection liberation. The most direct microscopic test is to calculate the effective transverse connection kernel and determine whether

$$M_A^2 = 0, \quad 0 < K_A < \infty$$

are dynamically realized. If the connection remains massive, algebraic, or purely composite, the construction yields internal geometric response but not a long-range gauge boson.

A tenth problem is the relation to gravitational response. The internal connection developed here belongs to a geometry-invisible multiplicity sector, while the spacetime Lorentz connection compares geometric frames. They are structurally distinct:

$$\omega : \text{geometric comparison}, \quad A : \text{multiplicity comparison}. \quad (232)$$

A complete MS quantum theory must determine how their quantum fluctuations, matter sources, and response kernels coexist without identifying their gauge groups or dynamical roles.

The separation in Eq. (232) also prevents a misleading form of unification. The present paper does not claim that internal gauge symmetry is a subgroup of the Lorentz group or that the Lorentz connection itself becomes a Yang–Mills connection for matter charge. Internal structure appears in the commutant and multiplicity data of the comparison theory, while Lorentz transport continues to act on the geometry-visible representation factors.

This distinction is compatible with a total connection on a tensor-product carrier,

$$\nabla = \nabla^{\text{geom}} \otimes I + I \otimes \nabla^{\text{int}}, \quad (233)$$

but such a direct-sum form expresses coexistence and compatibility, not identity of the two interactions.

The conceptual outcome of the paper can therefore be stated without invoking a specific phenomenological gauge model. The MS comparison framework provides a route in which internal quantum information is associated with coherent multiplicity left unresolved by the geometry-visible representation. Natural relational observables generate the full noncommutative multiplicity algebra. Smooth variation of protected multiplicities produces bundle geometry and holonomy. Requiring transport to preserve fusion converts a collection of sectorwise comparison phases into a candidate common internal symmetry. Only an additional infrared dynamical mechanism can liberate the resulting connection into a propagating gauge field.

The main conceptual progression is therefore

$$\begin{aligned} \text{geometry-visible representation} &\longrightarrow \text{geometry-invisible multiplicity,} \\ \text{multiplicity} &\longrightarrow \text{noncommutative relational information,} \\ \text{relational information} &\longrightarrow \text{inter-event internal comparison,} \\ \text{fusion-compatible comparison} &\longrightarrow \text{common internal gauge structure,} \\ \text{infrared response dynamics} &\longrightarrow \text{possible gauge propagation.} \end{aligned} \quad (234)$$

The distinction among these stages is as important as their connection. Representation multiplicity is not yet a gauge group; a frame group is not yet a holonomy group; holonomy is not yet a universal gauge principle; a principal connection is not yet a propagating gauge boson; and projective spin is not yet Fermi statistics.

Within these boundaries, the construction establishes a specific extension of the quantum-comparison program. Matter-like internal structure need not be introduced as an unrelated microscopic label at the beginning of the theory. It can instead be sought in the coherent relational multiplicity of projective comparison sectors and in the consistency conditions required to compare, transport, and fuse that multiplicity across spacetime.

A Recoupling Matrices and the Four-Carrier Relational Qubit

This appendix gives an explicit derivation of the four-carrier relational qubit used in Section 3. The purpose is to make the noncommutativity of different neutral-comparison questions independent of any implicit recoupling convention.

Consider four minimal projective comparison carriers,

$$\mathcal{H}_4 = (\mathbb{C}^2)^{\otimes 4}, \quad (235)$$

with the collective $SU(2)$ action

$$U_4(g) = u_{\frac{1}{2}}(g)^{\otimes 4}.$$

The invariant sector is

$$\mathcal{I}_4 = \text{Inv}_{SU(2)}(\mathcal{H}_4), \quad \dim \mathcal{I}_4 = 2. \quad (236)$$

For every ordered pair $i < j$, define the normalized singlet

$$|s_{ij}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle), \quad (237)$$

and the triplet basis

$$\begin{aligned} |t_{ij}^+\rangle &= |\uparrow_i \uparrow_j\rangle, \\ |t_{ij}^0\rangle &= \frac{1}{\sqrt{2}} (|\uparrow_i \downarrow_j\rangle + |\downarrow_i \uparrow_j\rangle), \\ |t_{ij}^-\rangle &= |\downarrow_i \downarrow_j\rangle. \end{aligned} \quad (238)$$

We use the (12)(34) coupling channel to define the logical basis

$$|0_L\rangle = |s_{12}\rangle |s_{34}\rangle, \quad (239)$$

and

$$|1_L\rangle = \frac{1}{\sqrt{3}} (|t_{12}^+ t_{34}^- \rangle - |t_{12}^0 t_{34}^0 \rangle + |t_{12}^- t_{34}^+ \rangle). \quad (240)$$

Both states have total spin zero and satisfy

$$\langle 0_L | 1_L \rangle = 0.$$

In the computational basis ordered as

$$|s_1 s_2 s_3 s_4\rangle, \quad s_i \in \{\uparrow, \downarrow\},$$

the first state is

$$|0_L\rangle = \frac{1}{2} (|\uparrow \downarrow \uparrow \downarrow\rangle - |\uparrow \downarrow \downarrow \uparrow\rangle - |\downarrow \uparrow \uparrow \downarrow\rangle + |\downarrow \uparrow \downarrow \uparrow\rangle). \quad (241)$$

The second is

$$\begin{aligned} |1_L\rangle &= \frac{1}{\sqrt{3}} \left[|\uparrow \uparrow \downarrow \downarrow\rangle + |\downarrow \downarrow \uparrow \uparrow\rangle \right. \\ &\quad \left. - \frac{1}{2} (|\uparrow \downarrow \uparrow \downarrow\rangle + |\uparrow \downarrow \downarrow \uparrow\rangle + |\downarrow \uparrow \uparrow \downarrow\rangle + |\downarrow \uparrow \downarrow \uparrow\rangle) \right]. \end{aligned} \quad (242)$$

The singlet projector of carriers 1 and 2 is

$$P_{12}^{(0)} = |s_{12}\rangle \langle s_{12}| \otimes I_{34}. \quad (243)$$

Since

$$|0_L\rangle$$

contains a 12-singlet while

$$|1_L\rangle$$

contains a 12-triplet,

$$\begin{aligned} P_{12}^{(0)}|0_L\rangle &= |0_L\rangle, \\ P_{12}^{(0)}|1_L\rangle &= 0. \end{aligned} \quad (244)$$

Hence

$$P_{12}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (245)$$

We next change the recoupling channel. The unique total singlet in which carriers 2 and 3 form a singlet is

$$|\phi_{23}\rangle = |s_{23}\rangle|s_{14}\rangle, \quad (246)$$

with the tensor factors restored to the order 1, 2, 3, 4. Explicitly,

$$|\phi_{23}\rangle = \frac{1}{2} (|\uparrow\uparrow\downarrow\downarrow\rangle - |\downarrow\uparrow\downarrow\uparrow\rangle - |\uparrow\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\uparrow\rangle). \quad (247)$$

Taking inner products with Eqs. (241) and (242) gives

$$\langle 0_L|\phi_{23}\rangle = -\frac{1}{2}, \quad \langle 1_L|\phi_{23}\rangle = \frac{\sqrt{3}}{2}. \quad (248)$$

Therefore

$$|\phi_{23}\rangle = -\frac{1}{2}|0_L\rangle + \frac{\sqrt{3}}{2}|1_L\rangle. \quad (249)$$

Since $P_{23}^{(0)}$ restricted to $\mathcal{I}_4$ projects onto this one-dimensional state,

$$P_{23}^{(0)}|_{\mathcal{I}_4} = |\phi_{23}\rangle\langle\phi_{23}|, \quad (250)$$

and hence

$$P_{23}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (251)$$

The third overlapping pairing gives a complementary recoupling direction. Let

$$|\phi_{13}\rangle$$

denote the normalized invariant state in which carriers 1 and 3 form a singlet. With a consistent singlet-orientation convention one obtains

$$|\phi_{13}\rangle = -\frac{1}{2}|0_L\rangle - \frac{\sqrt{3}}{2}|1_L\rangle \quad (252)$$

up to an irrelevant overall phase. Consequently,

$$P_{13}^{(0)}|_{\mathcal{I}_4} = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (253)$$

The three pairwise projectors therefore obey

$$P_{12}^{(0)} + P_{23}^{(0)} + P_{13}^{(0)} = \frac{3}{2}I_{I_4}. \quad (254)$$

This identity is the four-carrier singlet-sector form of the constraint imposed by total spin zero.

Indeed,

$$\mathbf{J}_{\text{tot}}^2 = \sum_{i=1}^4 \mathbf{J}_i^2 + 2 \sum_{i<j} \mathbf{J}_i \cdot \mathbf{J}_j. \quad (255)$$

For spin- $\frac{1}{2}$,

$$\mathbf{J}_i^2 = \frac{3}{4}I,$$

while

$$P_{ij}^{(0)} = \frac{1}{4}I - \mathbf{J}_i \cdot \mathbf{J}_j. \quad (256)$$

Thus pairwise singlet relations and pairwise rotational scalar products carry equivalent information.

Using Eqs. (245) and (251),

$$[P_{12}^{(0)}, P_{23}^{(0)}] = \frac{\sqrt{3}}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (257)$$

Hence

$$P_{12}^{(0)} \quad \text{and} \quad P_{23}^{(0)}$$

cannot be simultaneously diagonalized on the neutral relational qubit.

It is useful to express the three pairing questions as Bloch directions of the logical qubit. Define the logical Pauli matrices relative to

$$\{|0_L\rangle, |1_L\rangle\}.$$

Then

$$P_{12}^{(0)} = \frac{1}{2}(I + Z_L), \quad (258)$$

while

$$P_{23}^{(0)} = \frac{1}{2}I - \frac{\sqrt{3}}{4}X_L - \frac{1}{4}Z_L, \quad (259)$$

and

$$P_{13}^{(0)} = \frac{1}{2}I + \frac{\sqrt{3}}{4}X_L - \frac{1}{4}Z_L. \quad (260)$$

The corresponding three Bloch vectors lie in one great circle and are separated pairwise by 120° .

The three relational questions are therefore neither orthogonal projective measurements nor mutually commuting classical predicates. They form three different rank-one resolutions associated with different ways of recoupling the same four elementary projective carriers.

The relative angles follow directly from the overlaps. For example,

$$\text{Tr}(P_{12}^{(0)}P_{23}^{(0)}) = \frac{1}{4}. \quad (261)$$

For rank-one qubit projectors

$$P_{\mathbf{n}} = \frac{1}{2}(I + \mathbf{n} \cdot \boldsymbol{\sigma}),$$

one has

$$\text{Tr}(P_{\mathbf{n}}P_{\mathbf{m}}) = \frac{1}{2} (1 + \mathbf{n} \cdot \mathbf{m}).$$

Thus

$$\mathbf{n}_{12} \cdot \mathbf{n}_{23} = -\frac{1}{2},$$

which gives the 120° separation.

This geometry should not be confused with the physical spatial direction sphere of one spin carrier. The Bloch sphere here parametrizes pure states of the multiplicity qubit. Its coordinates describe coherent relational recouplings after total geometric spin has already been fixed to zero.

The matrices also permit a direct reconstruction of the logical Pauli algebra:

$$Z_L = 2P_{12}^{(0)} - I, \quad (262)$$

$$X_L = -\frac{2}{\sqrt{3}} (P_{23}^{(0)} - P_{13}^{(0)}), \quad (263)$$

$$Y_L = \frac{4i}{\sqrt{3}} [P_{12}^{(0)}, P_{23}^{(0)}]. \quad (264)$$

Consequently the physically natural recoupling projectors generate every operator on $\mathcal{I}_4$.

There is also a useful interpretation in terms of exchange operators. Let

$$S_{ij}$$

exchange the two carriers i and j . Since the exchange eigenvalue is -1 on the two-spin singlet and $+1$ on the triplet,

$$S_{ij} = I - 2P_{ij}^{(0)}. \quad (265)$$

On the relational qubit,

$$S_{12} = -Z_L, \quad (266)$$

whereas

$$S_{23} = \frac{1}{2}Z_L + \frac{\sqrt{3}}{2}X_L. \quad (267)$$

Thus permutations of the microscopic projective carriers act nontrivially on the relational multiplicity even though collective $SU(2)$ acts trivially.

This does not mean that physical exchange statistics have been derived. The operators S_{ij} here implement permutations of tensor factors in the composite carrier space. Identifying such permutations with adiabatic exchange of localized relativistic particles requires additional locality and statistics assumptions. Their role in the present appendix is purely representation theoretic.

The explicit calculation establishes three facts used in the main text:

$$\begin{aligned} \dim \mathcal{I}_4 &= 2, \\ [P_{12}^{(0)}, P_{23}^{(0)}] &\neq 0, \\ \text{Alg} \{P_{12}^{(0)}, P_{23}^{(0)}\} &= M_2(\mathbb{C}). \end{aligned} \quad (268)$$

The four-carrier relational qubit is therefore not merely a degeneracy space. Natural geometry-scalar comparison questions already endow it with the full noncommutative qubit observable algebra.

B Schur–Weyl and Temperley–Lieb Structure of the Relational Commutant

We now give a more general derivation of the relation between pairwise neutral-comparison observables and the full commutant of collective rotational comparison.

Let

$$\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N} \quad (269)$$

and let

$$U_N(g) = g^{\otimes N}, \quad g \in SU(2), \quad (270)$$

denote the collective fundamental action.

The symmetric group S_N acts on the same Hilbert space by permutation of tensor factors. For

$$\pi \in S_N,$$

write

$$U(\pi)(v_1 \otimes \cdots \otimes v_N) = v_{\pi^{-1}(1)} \otimes \cdots \otimes v_{\pi^{-1}(N)}. \quad (271)$$

Since the same matrix g acts on every tensor factor,

$$[U(\pi), U_N(g)] = 0 \quad (272)$$

for every

$$\pi \in S_N, \quad g \in SU(2).$$

Thus the image of the group algebra

$$\mathbb{C}[S_N]$$

lies in the collective commutant.

Conversely, Schur–Weyl duality gives the decomposition

$$(\mathbb{C}^2)^{\otimes N} \simeq \bigoplus_{\lambda \vdash N, \ell(\lambda) \leq 2} V_\lambda^{U(2)} \otimes W_\lambda^{S_N}, \quad (273)$$

where λ runs over partitions of N with at most two rows.

Restricting the $U(2)$ factor to $SU(2)$ gives the total-spin decomposition

$$\mathcal{H}_N \simeq \bigoplus_j V_j \otimes M_j^{(N)}. \quad (274)$$

The two-row partition associated with spin j is

$$\lambda_j = \left(\frac{N}{2} + j, \frac{N}{2} - j \right). \quad (275)$$

The corresponding symmetric-group representation provides the multiplicity space:

$$M_j^{(N)} \simeq W_{\lambda_j}^{S_N}. \quad (276)$$

The restriction from $U(2)$ to $SU(2)$ does not enlarge the commutant in this fixed- N tensor power. The central $U(1) \subset U(2)$ acts as an overall scalar on $(\mathbb{C}^2)^{\otimes N}$, and therefore the $SU(2)$ and $U(2)$ collective commutants coincide.

Consequently,

$$\text{End}_{SU(2)}(\mathcal{H}_N) = \text{im}(\mathbb{C}[S_N] \longrightarrow \text{End}(\mathcal{H}_N)). \quad (277)$$

For $N > 2$, this representation of the full group algebra is not faithful; only the two-row representation sectors survive. Thus it is the image, or equivalently the appropriate quotient of $\mathbb{C}[S_N]$, that equals the collective commutant.

The pairwise singlet projectors provide a particularly natural generating set for this image. Let

$$S_{ij}$$

denote the transposition of carriers i and j . As reviewed in Appendix A,

$$S_{ij} = I - 2P_{ij}^{(0)}. \quad (278)$$

Therefore

$$P_{ij}^{(0)} = \frac{1}{2}(I - S_{ij}). \quad (279)$$

Since transpositions generate S_N , the unital algebra generated by all pairwise singlet projectors contains the entire image of $\mathbb{C}[S_N]$. The reverse inclusion follows because every $P_{ij}^{(0)}$ commutes with collective $SU(2)$.

This proves the main algebraic statement used in Section 3.

Theorem B.1 (Pairwise relational centralizer theorem). *Let*

$$\mathcal{R}_N = \text{Alg}\{P_{ij}^{(0)}; 1 \leq i < j \leq N\}. \quad (280)$$

Then

$$\mathcal{R}_N = \text{End}_{SU(2)}((\mathbb{C}^2)^{\otimes N}). \quad (281)$$

Under the total-spin decomposition,

$$\mathcal{R}_N \simeq \bigoplus_j I_{V_j} \otimes M_{m_j^{(N)}}(\mathbb{C}). \quad (282)$$

Proof. Every $P_{ij}^{(0)}$ is an $SU(2)$ -scalar, so

$$\mathcal{R}_N \subseteq \text{End}_{SU(2)}(\mathcal{H}_N).$$

Conversely, Eq. (278) implies that $\mathcal{R}_N$ contains every transposition. Since transpositions generate the permutation group, $\mathcal{R}_N$ contains the image of $\mathbb{C}[S_N]$. Schur–Weyl duality identifies this image with the full collective commutant, proving Eq. (281).

The block form follows from Schur’s lemma applied to Eq. (274). $\square$

The theorem has a direct interpretation in the MS comparison setting. Once the total geometry-visible rotational representation is fixed, every operator acting on the multiplicity space can be synthesized algebraically from pairwise neutral-comparison observables.

The result does not assert that every element of the commutant is equally simple dynamically. A high-degree polynomial in pairwise projectors may correspond to a complicated many-body operation. The theorem is an algebraic generation statement rather than a claim of uniform dynamical accessibility.

A smaller generating set is sufficient. Let

$$e_i := 2P_{i,i+1}^{(0)}, \quad i = 1, \dots, N-1. \quad (283)$$

Equivalently,

$$e_i = I - S_{i,i+1}. \quad (284)$$

The adjacent transpositions generate S_N , so the e_i generate $\mathcal{R}_N$.

For the spin- $\frac{1}{2}$ representation they obey

$$e_i^2 = 2e_i, \quad (285)$$

$$e_i e_{i+1} e_i = e_i, \quad (286)$$

$$e_{i+1} e_i e_{i+1} = e_{i+1}, \quad (287)$$

$$[e_i, e_j] = 0, \quad |i - j| > 1. \quad (288)$$

These are the defining relations of the Temperley–Lieb algebra at loop parameter

$$\delta = 2.$$

For completeness, the first relation follows immediately from the idempotence of $P_{i,i+1}^{(0)}$:

$$e_i^2 = 4P_{i,i+1}^{(0)} = 2e_i.$$

The distant-commutativity relation follows because projectors acting on disjoint tensor factors commute.

The nontrivial adjacent relation can be verified on three consecutive spin- $\frac{1}{2}$ carriers. Since

$$P_{12}^{(0)}$$

projects the first two carriers to a one-dimensional singlet, recoupling through a 23-singlet and back produces

$$P_{12}^{(0)} P_{23}^{(0)} P_{12}^{(0)} = \frac{1}{4} P_{12}^{(0)}. \quad (289)$$

Multiplying by 2^3 gives

$$e_1 e_2 e_1 = e_1.$$

The general adjacent relation follows identically.

Thus the collective commutant is the finite spin-chain representation of

$$TL_N(2).$$

Proposition B.2 (Temperley–Lieb realization of relational comparison). *The relational algebra generated by adjacent neutral-pair observables*

$$2P_{i,i+1}^{(0)}$$

is the image of the Temperley–Lieb algebra $TL_N(2)$ on

$$(\mathbb{C}^2)^{\otimes N}.$$

This image equals the full commutant of collective $SU(2)$.

The Temperley–Lieb description makes the local structure of relational incompatibility transparent. If

$$|i - j| > 1,$$

the two neutral-pair questions concern disjoint pairs and commute. If the pairs overlap, the corresponding operators are generally noncommuting but satisfy the constrained recoupling relation

$$e_i e_{i+1} e_i = e_i.$$

The structure therefore lies between two extremes: the relational observables are neither a commuting family of classical pair labels nor a free matrix algebra with no constraints. Their noncommutativity is organized by recoupling.

The dimension of the complete collective commutant is

$$\dim_{\mathbb{C}} \mathcal{R}_N = \sum_j \left(m_j^{(N)}\right)^2. \quad (290)$$

For the fundamental $SU(2)$ tensor power, this sum is the Catalan number

$$\dim_{\mathbb{C}} \mathcal{R}_N = C_N = \frac{1}{N+1} \binom{2N}{N}. \quad (291)$$

One way to see this is through the Temperley–Lieb realization: the abstract algebra $TL_N(2)$ has a basis indexed by planar noncrossing pairings of $2N$ boundary points, whose number is C_N . The standard spin- $\frac{1}{2}$ representation realizes the corresponding centralizer algebra.

Equivalently, using

$$m_j^{(N)} = \binom{N}{\frac{N}{2} - j} - \binom{N}{\frac{N}{2} - j - 1},$$

one obtains the combinatorial identity

$$\sum_j \left[\binom{N}{\frac{N}{2} - j} - \binom{N}{\frac{N}{2} - j - 1} \right]^2 = \frac{1}{N+1} \binom{2N}{N}. \quad (292)$$

The first few dimensions are

$$\dim \mathcal{R}_N = 1, 2, 5, 14, 42, 132, \dots \quad (293)$$

for

$$N = 1, 2, 3, 4, 5, 6, \dots$$

For example,

$$N = 4$$

gives

$$1^2 + 3^2 + 2^2 = 14,$$

in agreement with

$$C_4 = 14.$$

Its large- N asymptotic form is

$$\dim \mathcal{R}_N \sim \frac{4^N}{\sqrt{\pi} N^{3/2}}. \quad (294)$$

Hence

$$\log_2 \dim \mathcal{R}_N = 2N - \frac{3}{2} \log_2 N + O(1). \quad (295)$$

This scaling concerns the dimension of an operator algebra and should not be identified with the number of encoded logical qubits. It nevertheless shows that the algebra invisible to collective geometry remains exponentially large.

There are therefore two distinct Catalan structures in the present construction.

For an even number

$$N = 2n$$

of carriers, the dimension of the total-singlet state space is

$$\dim \mathcal{I}_{2n} = C_n. \quad (296)$$

By contrast, the dimension of the full collective commutant for N carriers is

$$\dim \mathcal{R}_N = C_N. \quad (297)$$

The first counts neutral relational states; the second counts independent collective-geometry-invariant operators across all total-spin sectors.

This distinction is important when interpreting the internal structure. The neutral sector

$$\mathcal{I}_{2n}$$

is one particular multiplicity block, whereas the full relational commutant contains

$$\text{End}(M_j^{(N)})$$

for every allowed total spin j .

The permutation formulation also clarifies why the multiplicity information is relational. A permutation changes the coupling pattern of constituent carriers while commuting with the collective action

$$g^{\otimes N}.$$

Thus the same total geometric transformation is compatible with different internal arrangements of the constituents.

At the same time, permutation symmetry should not be identified with an internal gauge group. The finite symmetric group acts on the microscopic tensor factors and generates the commutant algebra, whereas the spacetime internal group studied in Sections 4 and 5 is reconstructed from smooth inter-event transport and tensor-compatible sector automorphisms. These are different levels of structure.

The finite-dimensional result established here is

$$\begin{aligned} \text{collective } SU(2) &\longleftrightarrow \text{geometry-visible spin,} \\ \mathbb{C}[S_N] \text{ image} &\longleftrightarrow \text{geometry-invisible multiplicity algebra,} \\ TL_N(2) &\longleftrightarrow \text{local neutral-recoupling generators.} \end{aligned} \quad (298)$$

In this precise sense, the relational commutant does not need to be added to the composite theory as an independent microscopic matrix algebra. It is already generated by the algebra of neutral relations among the elementary projective comparison carriers.

C Projector Geometry and the Induced Relational Curvature

This appendix derives the projector and frame formulas for the induced connection used in Section 4. The derivation is included in order to separate three structures that are easily conflated: the connection of the ambient comparison bundle, the motion of the protected multiplicity subspace inside that bundle, and the gauge freedom associated with the choice of a local multiplicity frame.

Let

$$\mathcal{R} \longrightarrow M \quad (299)$$

be a rank- N Hermitian vector bundle equipped with a Hermitian connection

$$D : \Gamma(\mathcal{R}) \longrightarrow \Omega^1(M, \mathcal{R}), \quad (300)$$

with curvature

$$F_{\mathcal{R}} = D^2. \quad (301)$$

Let

$$\mathcal{M} \subset \mathcal{R}$$

be a rank- m Hermitian subbundle. On a local trivializing patch choose an orthonormal frame

$$Z = (|m_1\rangle, \dots, |m_m\rangle), \quad Z^\dagger Z = I_m. \quad (302)$$

Fiberwise,

$$Z : \mathbb{C}^m \rightarrow \mathcal{R}_x$$

is an isometric embedding.

The orthogonal projector onto the selected multiplicity subspace is

$$P = ZZ^\dagger. \quad (303)$$

It satisfies

$$P^2 = P, \quad P^\dagger = P. \quad (304)$$

Differentiating

$$P^2 = P$$

covariantly gives

$$(DP)P + P(DP) = DP. \quad (305)$$

Multiplying from the left and right by P yields

$$P(DP)P = 0. \quad (306)$$

Similarly,

$$(I - P)(DP)(I - P) = 0. \quad (307)$$

Thus DP is purely off diagonal with respect to the decomposition

$$\mathcal{R} = \mathcal{M} \oplus \mathcal{M}^\perp. \quad (308)$$

The induced connection on $\mathcal{M}$ is

$$\nabla^{\mathcal{M}} = PD|_{\Gamma(\mathcal{M})}. \quad (309)$$

If

$$\psi = Zc$$

is a local section of $\mathcal{M}$, then

$$\begin{aligned} PD(Zc) &= P(DZc + Zdc) \\ &= Z(Z^\dagger DZc + dc). \end{aligned} \quad (310)$$

Hence the local connection matrix is

$$A = Z^\dagger DZ. \quad (311)$$

Differentiating the orthonormality condition

$$Z^\dagger Z = I_m$$

gives

$$(DZ)^\dagger Z + Z^\dagger DZ = 0. \quad (312)$$

Therefore

$$A^\dagger = -A, \quad (313)$$

as required for a $U(m)$ connection.

Define the normal or extrinsic derivative

$$K := (I - P)DZ. \quad (314)$$

Then

$$DZ = ZA + K. \quad (315)$$

By construction,

$$Z^\dagger K = 0. \quad (316)$$

The same object can be expressed directly in terms of the projector.

Since

$$PZ = Z,$$

covariant differentiation gives

$$(DP)Z + PDZ = DZ. \quad (317)$$

Therefore

$$(DP)Z = (I - P)DZ = K. \quad (318)$$

Since DP is Hermitian as an endomorphism-valued one-form,

$$Z^\dagger DP = K^\dagger. \quad (319)$$

Thus K and DP contain the same information about the motion of the selected subspace.

We now derive the curvature formula.

Theorem C.1 (Projector curvature formula). *The curvature of the induced connection*

$$\nabla^{\mathcal{M}} = PD$$

is

$$F_{\mathcal{M}} = PF_{\mathcal{R}}P + P(DP \wedge DP)P. \quad (320)$$

Proof. Let

$$s \in \Gamma(\mathcal{M}), \quad Ps = s.$$

Then

$$\begin{aligned} (\nabla^{\mathcal{M}})^2 s &= PD(PDs) \\ &= P(DP) \wedge Ds + PD^2 s. \end{aligned} \quad (321)$$

Differentiating

$$Ps = s$$

gives

$$(DP)s + PDs = Ds,$$

and hence

$$(I - P)Ds = (DP)s. \quad (322)$$

Using

$$P(DP)P = 0,$$

only the normal component of Ds contributes to the first term of Eq. (321). Therefore

$$\begin{aligned} P(DP) \wedge Ds &= P(DP) \wedge (I - P)Ds \\ &= P(DP \wedge DP)s. \end{aligned} \quad (323)$$

The second term gives

$$PD^2 s = PF_{\mathcal{R}}Ps.$$

Combining the two expressions proves Eq. (320). $\square$

In a local multiplicity frame, the same result takes a Gauss-type form.

Theorem C.2 (Frame form of the induced curvature). *Let*

$$A = Z^\dagger DZ, \quad K = (I - P)DZ.$$

Then

$$F_A = dA + A \wedge A = Z^\dagger F_{\mathcal{R}}Z + K^\dagger \wedge K. \quad (324)$$

Proof. Starting from

$$A = Z^\dagger DZ,$$

one obtains

$$dA = (DZ)^\dagger \wedge DZ + Z^\dagger F_{\mathcal{R}}Z. \quad (325)$$

Using

$$DZ = ZA + K$$

and

$$(DZ)^\dagger Z = -A, \quad Z^\dagger K = 0,$$

gives

$$(DZ)^\dagger \wedge DZ = -A \wedge A + K^\dagger \wedge K. \quad (326)$$

Substitution into Eq. (325) yields Eq. (324). $\square$

Equations (320) and (324) are equivalent. Indeed,

$$K = (DP)Z$$

implies

$$K^\dagger \wedge K = Z^\dagger (DP \wedge DP)Z. \quad (327)$$

If the ambient comparison connection is flat,

$$F_{\mathcal{R}} = 0, \quad (328)$$

then

$$F_{\mathcal{M}} = P(DP \wedge DP)P, \quad (329)$$

and

$$F_A = K^\dagger \wedge K. \quad (330)$$

Thus the complete induced curvature is generated by the extrinsic motion of the physical relational subspace.

In a trivial ambient bundle with

$$D = d,$$

this becomes

$$F_{\mathcal{M}} = P(dP \wedge dP)P. \quad (331)$$

This is the standard universal Grassmannian curvature formula.

The frame freedom leaves the projector invariant. If

$$Z' = ZU, \quad U : M \rightarrow U(m), \quad (332)$$

then

$$P' = P,$$

while

$$A' = U^{-1}AU + U^{-1}dU \quad (333)$$

and

$$F'_A = U^{-1}F_A U. \quad (334)$$

The extrinsic derivative transforms as

$$K' = KU. \quad (335)$$

The curvature is therefore an intrinsic property of the selected subbundle, even though its matrix representation depends on the local multiplicity frame.

A particularly important special case occurs when the physical subspace is fixed while its frame changes.

Proposition C.3 (Constant-projector pure-gauge result). *Let the ambient bundle be flat and locally trivialized so that $D = d$. If*

$$P(x) = P_0$$

is constant on a simply connected patch, then every local orthonormal frame has the form

$$Z(x) = Z_0 U(x), \quad (336)$$

with

$$U(x) \in U(m),$$

and

$$A = U^{-1}dU, \quad F_A = 0. \quad (337)$$

This proposition identifies the precise content of the statement that multiplicity dimension alone does not generate an internal field strength. The existence of the local frame freedom

$$U(m)$$

is kinematical. Nonzero curvature requires additional geometric data.

In the flat ambient case,

$$F_A \neq 0 \implies dP \neq 0. \quad (338)$$

The converse does not hold in general, because a nonconstant map into the Grassmannian may have vanishing pulled-back curvature on particular two-dimensional tangent directions.

The local variation of the protected subspace can be quantified by a matrix-valued quantum geometric tensor. In local coordinates

$$x^\mu,$$

define

$$K_\mu = (I - P)D_\mu Z. \quad (339)$$

Then

$$Q_{\mu\nu} = K_\mu^\dagger K_\nu. \quad (340)$$

The extrinsic contribution to curvature is

$$F_{\mu\nu}^{\text{ext}} = Q_{\mu\nu} - Q_{\nu\mu}. \quad (341)$$

Define the scalar positive tensor

$$g_{\mu\nu}^{\text{rel}} = \frac{1}{2} \text{tr} (Q_{\mu\nu} + Q_{\nu\mu}). \quad (342)$$

In particular,

$$g_{\mu\mu}^{\text{rel}} = \text{tr} (K_\mu^\dagger K_\mu) = \|K_\mu\|_{\text{HS}}^2. \quad (343)$$

Using

$$\|AB\|_{\text{HS}} \leq \|A\|_{\text{HS}} \|B\|_{\text{HS}},$$

one obtains

$$\begin{aligned} \|F_{\mu\nu}^{\text{ext}}\|_{\text{HS}} &\leq \|K_\mu^\dagger K_\nu\|_{\text{HS}} + \|K_\nu^\dagger K_\mu\|_{\text{HS}} \\ &\leq 2\|K_\mu\|_{\text{HS}} \|K_\nu\|_{\text{HS}}. \end{aligned} \quad (344)$$

Therefore

$$\|F_{\mu\nu}^{\text{ext}}\|_{\text{HS}} \leq 2 \sqrt{g_{\mu\mu}^{\text{rel}} g_{\nu\nu}^{\text{rel}}}. \quad (345)$$

The inequality gives a direct relation between local subspace motion and the maximum size of the curvature generated by that motion.

The projector formalism also encodes topology. For a rank- m complex multiplicity bundle, the standard Chern–Weil representatives are constructed from the induced curvature. With the anti-Hermitian convention used here, the first Chern form is

$$c_1(\mathcal{M}) = \frac{i}{2\pi} \operatorname{tr} F_A, \quad (346)$$

while the second Chern character is represented by

$$\operatorname{ch}_2(\mathcal{M}) = \frac{1}{2} \left(\frac{i}{2\pi} \right)^2 \operatorname{tr} (F_A \wedge F_A). \quad (347)$$

In a flat ambient trivialization these invariants are entirely determined by the map

$$P : M \rightarrow \operatorname{Gr}(m, N).$$

They therefore measure global twisting of the protected relational subspace, not merely local frame choice.

Their existence should nevertheless be interpreted only as a topological capacity statement. The present construction does not imply that arbitrary Chern classes are dynamically realized by an MS comparison vacuum.

The main identities established in this appendix are

$$\begin{aligned} P &= ZZ^\dagger, \\ A &= Z^\dagger DZ, \\ K &= (I - P)DZ = (DP)Z, \\ F_{\mathcal{M}} &= PF_{\mathcal{R}}P + P(DP \wedge DP)P, \\ F_A &= Z^\dagger F_{\mathcal{R}}Z + K^\dagger \wedge K. \end{aligned} \quad (348)$$

They make explicit why a moving relational subspace can possess nontrivial internal comparison curvature even when the ambient comparison bundle is flat, while a mere basis transformation inside a fixed subspace cannot.

D Tensor-Natural Transport and the Principal Comparison Bundle

This appendix gives a more explicit formulation of the conditional principal-bundle reconstruction used in Section 5. The purpose is not to reproduce the general Tannaka–Krein or Doplicher–Roberts reconstruction theorems, but to show precisely how tensor-natural inter-event comparison produces a principal bundle once the required categorical hypotheses are assumed.

Let

$$\mathcal{C}_{\text{rel}}$$

be a rigid unitary tensor category of localized relational sectors. We assume a simple tensor unit

$$\mathbf{1},$$

finite-dimensional morphism spaces, finite direct sums, subobjects, and conjugates. Whenever reconstruction by an ordinary compact group is desired, we additionally assume the symmetric tensor structure required by the corresponding reconstruction theorem.

At every event $x \in M$, let

$$\omega_x : C_{\text{rel}} \longrightarrow \text{Hilb}_{\text{fd}} \quad (349)$$

be a faithful unitary tensor functor. The multiplicity fiber of a sector r is

$$M_r(x) = \omega_x(r). \quad (350)$$

Choose a reference event

$$x_0 \in M$$

and write

$$\omega_0 := \omega_{x_0}. \quad (351)$$

Define

$$G := \text{Aut}^{\otimes}(\omega_0). \quad (352)$$

An element

$$g \in G$$

is a family

$$g = \{g_r\}_r$$

with

$$g_r \in U(\omega_0(r))$$

such that

$$g_s \omega_0(f) = \omega_0(f) g_r \quad (353)$$

for every morphism

$$f : r \rightarrow s,$$

and

$$g_{r \otimes s} = g_r \otimes g_s, \quad g_1 = I. \quad (354)$$

For every sector r , evaluation gives a unitary representation

$$\rho_r : G \longrightarrow U(\omega_0(r)), \quad \rho_r(g) = g_r. \quad (355)$$

At a general event x , define

$$\mathcal{P}_x := \text{Iso}^{\otimes}(\omega_0, \omega_x), \quad (356)$$

the set of unitary tensor-natural isomorphisms

$$p : \omega_0 \Rightarrow \omega_x.$$

Thus an element

$$p \in \mathcal{P}_x$$

is a family

$$p = \{p_r\}_r, \quad p_r : \omega_0(r) \longrightarrow \omega_x(r),$$

satisfying

$$p_s \omega_0(f) = \omega_x(f) p_r \quad (357)$$

and

$$p_{r \otimes s} = p_r \otimes p_s, \quad p_1 = I. \quad (358)$$

The group G acts on $\mathcal{P}_x$ on the right by

$$p \cdot g := p \circ g. \quad (359)$$

Sectorwise,

$$(p \cdot g)_r = p_r g_r. \quad (360)$$

Proposition D.1 (Tensor-isomorphism torsor). *If $\mathcal{P}_x \neq \emptyset$, then $\mathcal{P}_x$ is a right G -torsor.*

Proof. Let

$$p, q \in \mathcal{P}_x.$$

Then

$$p^{-1}q : \omega_0 \Rightarrow \omega_0$$

is a unitary tensor-natural automorphism, so

$$p^{-1}q \in G.$$

Hence

$$q = p \cdot (p^{-1}q),$$

which proves transitivity.

If

$$p \cdot g = p,$$

then composition with p^{-1} gives

$$g = I.$$

The action is therefore free. □

Suppose now that the family

$$\{\omega_x\}_{x \in M}$$

varies smoothly in the sense that the torsors $\mathcal{P}_x$ admit smooth local sections and assemble into a smooth total space

$$\mathcal{P} = \bigsqcup_{x \in M} \mathcal{P}_x. \quad (361)$$

Then

$$\pi : \mathcal{P} \longrightarrow M \quad (362)$$

is a principal G -bundle.

Theorem D.2 (Conditional principal comparison bundle). *Assume that:*

1. every $\mathcal{P}_x$ is nonempty;
2. the family of tensor fiber functors ω_x varies smoothly;
3. the torsors $\mathcal{P}_x$ admit smooth local sections.

Then

$$\mathcal{P} = \bigsqcup_x \text{Iso}^\otimes(\omega_0, \omega_x)$$

is a principal bundle over M with structure group

$$G = \text{Aut}^\otimes(\omega_0).$$

The associated vector bundle for a sector r can be reconstructed directly from $\mathcal{P}$. Let

$$V_r := \omega_0(r). \quad (363)$$

Using the representation

$$\rho_r(g) = g_r,$$

form

$$\mathcal{E}_r := \mathcal{P} \times_{\rho_r} V_r. \quad (364)$$

We use the convention

$$(p \cdot g, v) \sim (p, \rho_r(g)v). \quad (365)$$

Then define

$$\Phi_r : \mathcal{E}_r \longrightarrow M_r, \quad [p, v] \longmapsto p_r(v). \quad (366)$$

This is well defined because

$$\begin{aligned} \Phi_r([p \cdot g, v]) &= (p_r g_r)v \\ &= p_r(\rho_r(g)v) \\ &= \Phi_r([p, \rho_r(g)v]). \end{aligned} \quad (367)$$

Proposition D.3 (Sector bundles as associated bundles). *For every relational sector r ,*

$$M_r \simeq \mathcal{P} \times_{\rho_r} V_r \quad (368)$$

as Hermitian vector bundles.

Thus the apparently separate multiplicity bundles are all associated bundles of one principal comparison structure.

We now incorporate path comparison.

For a path

$$\gamma : x \rightarrow y,$$

suppose the relational comparison theory provides a tensor-natural unitary isomorphism

$$U_\gamma : \omega_x \Rightarrow \omega_y. \quad (369)$$

Sectorwise,

$$U_\gamma^{(r)} : M_r(x) \longrightarrow M_r(y). \quad (370)$$

Tensor naturality requires

$$U_\gamma^{(s)} \omega_x(f) = \omega_y(f) U_\gamma^{(r)} \quad (371)$$

for every

$$f : r \rightarrow s,$$

while monoidality gives

$$U_\gamma^{(r \otimes s)} = U_\gamma^{(r)} \otimes U_\gamma^{(s)}. \quad (372)$$

Assume also

$$U_{\gamma_2 \circ \gamma_1} = U_{\gamma_2} U_{\gamma_1}, \quad (373)$$

and

$$U_{\gamma^{-1}} = U_\gamma^{-1}. \quad (374)$$

A path $\gamma : x \rightarrow y$ induces a map

$$\mathcal{T}_\gamma : \mathcal{P}_x \longrightarrow \mathcal{P}_y, \quad \mathcal{T}_\gamma(p) = U_\gamma \circ p. \quad (375)$$

This map is G -equivariant:

$$\begin{aligned} \mathcal{T}_\gamma(p \cdot g) &= U_\gamma \circ p \circ g \\ &= \mathcal{T}_\gamma(p) \cdot g. \end{aligned} \quad (376)$$

Therefore tensor-natural comparison transport acts directly as principal parallel transport.

Proposition D.4 (Principal transport from tensor-natural comparison). *A family U_γ satisfying Eqs. (371)–(374) defines G -equivariant parallel transport on*

$$\mathcal{P} \rightarrow M.$$

Under the standard smoothness and locality hypotheses relating smooth parallel transport to connections, it determines a principal connection on $\mathcal{P}$.

The qualification concerning smoothness is important. An arbitrary assignment of unitary operators to paths need not arise from a smooth local connection. One requires the usual regular dependence on paths, locality, and compatibility under subdivision and reparametrization.

Choose a local section

$$s : U \subset M \longrightarrow \mathcal{P}. \quad (377)$$

The principal connection is represented locally by a Lie-algebra-valued one-form

$$A \in \Omega^1(U, \mathfrak{g}), \quad \mathfrak{g} = \text{Lie}(G). \quad (378)$$

For sector r , differentiation of

$$\rho_r : G \rightarrow U(V_r)$$

gives

$$\rho_{r*} : \mathfrak{g} \longrightarrow \mathfrak{u}(V_r). \quad (379)$$

The induced local connection is

$$A^{(r)} = \rho_{r*}(A). \quad (380)$$

Similarly, if

$$F = dA + A \wedge A \quad (381)$$

is the principal curvature, then

$$F^{(r)} = \rho_{r*}(F). \quad (382)$$

Thus the connections and curvatures observed in different relational sectors are not independent data.

The fusion relations follow automatically from tensor naturality. Let

$$C_{rs}^t : r \otimes s \rightarrow t$$

be a morphism in C_{rel} . Since

$$g \in \text{Aut}^\otimes(\omega_0)$$

is natural,

$$\rho_t(g)C_{rs}^t = C_{rs}^t(\rho_r(g) \otimes \rho_s(g)). \quad (383)$$

Differentiation gives

$$\rho_{t*}(X)C_{rs}^t = C_{rs}^t(\rho_{r*}(X) \otimes I + I \otimes \rho_{s*}(X)) \quad (384)$$

for every

$$X \in \mathfrak{g}.$$

Consequently the principal connection automatically satisfies

$$\nabla C_{rs}^t = 0 \quad (385)$$

when the fusion morphisms are represented in tensor-compatible local frames.

The same is true for parallel transport:

$$U_\gamma^{(t)} C_{rs}^t(x) = C_{rs}^t(y) (U_\gamma^{(r)} \otimes U_\gamma^{(s)}). \quad (386)$$

Thus the fusion-transport equation used in the main text is not an independent property after principal tensor reconstruction; it is built into the naturality of the common comparison transport.

Closed paths give a particularly direct relation between principal holonomy and tensor automorphisms. For a loop

$$\gamma : x_0 \rightarrow x_0,$$

one has

$$U_\gamma \in \text{Aut}^\otimes(\omega_0) = G. \quad (387)$$

Therefore

$$\text{Hol}_{x_0}(\mathcal{P}, A) \subseteq G. \quad (388)$$

In sector r ,

$$U_\gamma^{(r)} = \rho_r(U_\gamma). \quad (389)$$

Hence all sector holonomies are representation images of the same principal holonomy element.

This supplies a practical universality test. If a microscopic comparison model produces candidate sector holonomies

$$U_\gamma^{(r)},$$

then they can arise from one internal principal structure only if, for every loop and every fusion morphism, they form a tensor-natural family.

A collection of unrelated unitary matrices, even if each separately resembles a Berry holonomy, does not satisfy this criterion in general.

The choice of the reference event is not physical. Let

$$x_1$$

be another reference event and suppose

$$q : \omega_0 \Rightarrow \omega_1$$

is a tensor-natural unitary isomorphism. Then

$$\text{Aut}^\otimes(\omega_1) = q \text{Aut}^\otimes(\omega_0) q^{-1}. \quad (390)$$

Thus different choices of reference fiber functor produce canonically isomorphic internal groups, up to the expected conjugation associated with frame choice.

Likewise, a local change of principal section

$$s(x) \mapsto s(x)g(x) \quad (391)$$

induces the standard gauge transformation

$$A \mapsto g^{-1}Ag + g^{-1}dg. \quad (392)$$

In sector r ,

$$A^{(r)} \mapsto \rho_r(g)^{-1}A^{(r)}\rho_r(g) + \rho_r(g)^{-1}d\rho_r(g). \quad (393)$$

The ordinary gauge transformation law therefore appears only after the tensor-compatible family of multiplicity fibers has been organized into a single principal bundle. Before that reconstruction, independent local basis changes in separate multiplicity spaces need not correspond to one common gauge transformation.

The assumptions entering this construction should be kept explicit. A rigid unitary tensor category with a faithful fiber functor need not automatically be the representation category of an ordinary compact group unless the appropriate symmetry conditions are satisfied. If the comparison category is braided but not symmetric, or if the relevant fiber structure is more general, the reconstructed object may be a quantum group, weak Hopf algebra, or another generalized symmetry structure rather than an ordinary compact gauge group.

Accordingly,

$$\text{relational fusion category} \not\Rightarrow \text{ordinary compact gauge group} \quad (394)$$

without the additional reconstruction hypotheses used in the main text.

The principal-bundle construction also does not imply gauge-field dynamics. The bundle

$$\mathcal{P} \rightarrow M$$

and its connection encode how relational fibers are compared. Whether the connection possesses an independent massless transverse mode is the separate connection-liberation problem discussed in Section 6.

The logical role of the present appendix is therefore limited but precise. Under the stated tensor and smoothness assumptions,

$$\begin{aligned}
\{\omega_x\} &\longrightarrow \mathcal{P}_x = \text{Iso}^\otimes(\omega_0, \omega_x), \\
G &= \text{Aut}^\otimes(\omega_0), \\
\mathcal{P} &= \bigsqcup_x \mathcal{P}_x, \\
U_\gamma &\longrightarrow \text{principal parallel transport}, \\
A &\longrightarrow A^{(r)} = \rho_{r*}(A).
\end{aligned} \tag{395}$$

Thus tensor-natural comparison converts a family of local relational multiplicity spaces into one common principal comparison geometry.